

Theoretical Ecotoxicology

Session 5: Mixture and multiple stressor modelling

University Duisburg-Essen SS 2025

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Intended learning outcomes of today's lecture

- Understanding and applying the major concepts for mixture toxicity estimation
- Knowledge on the predictive power of mixture models
- Understanding the conceptual basis of multiple stressor effects and null models

Learning targets and study questions

- Understanding and applying the major concepts for mixture toxicity
 - What is the *something from nothing* effect?
 - Explain the concepts of IA, CA and sum TU.
 - Discuss the choice between IA and CA.
 - What are antagonism and synergism?
- Knowledge on the predictive power of mixture models
 - Outline shortcomings of CA and IA.
 - How well do CA and IA match with observations, and how strong is the deviation?

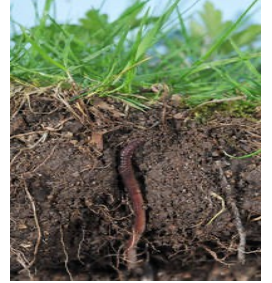
Learning targets and study questions

- Understanding the conceptual basis of multiple stressor effects and of null models
 - Explain why the choice of the null model matters for the evaluation of synergism and antagonism.
 - Describe some commonly used null models.
 - Which criteria can support the choice of the null model?

Mixtures abundant in ecosystems

76 of 82 target chemicals found in soils around Paris

Gaspéri et al. 2022 *Environ. Sci. Poll. Res.* 25: 23559



www.fao.org



www.wikipedia.org

398 of 2316 target chemicals found in Greek waste water samples

Gago-Ferrero et al. 2020 *J. Haz. Mat.* 387: 121712

47 of 92 target pesticides found on insects in Malaise traps in German conservation areas

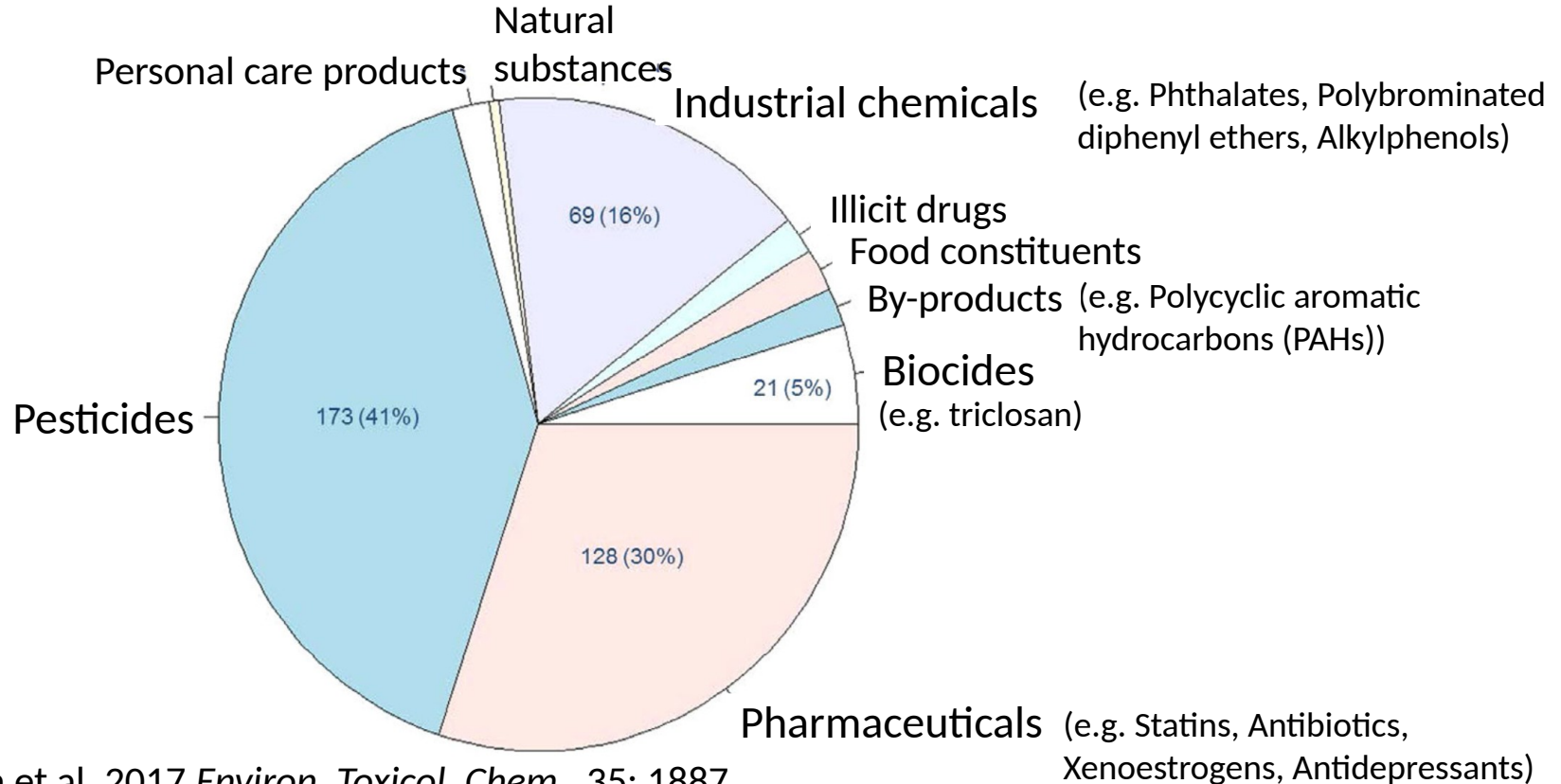
Brühl et al. 2021 *Sci. Rep.* 11: 24144



www.wikipedia.org

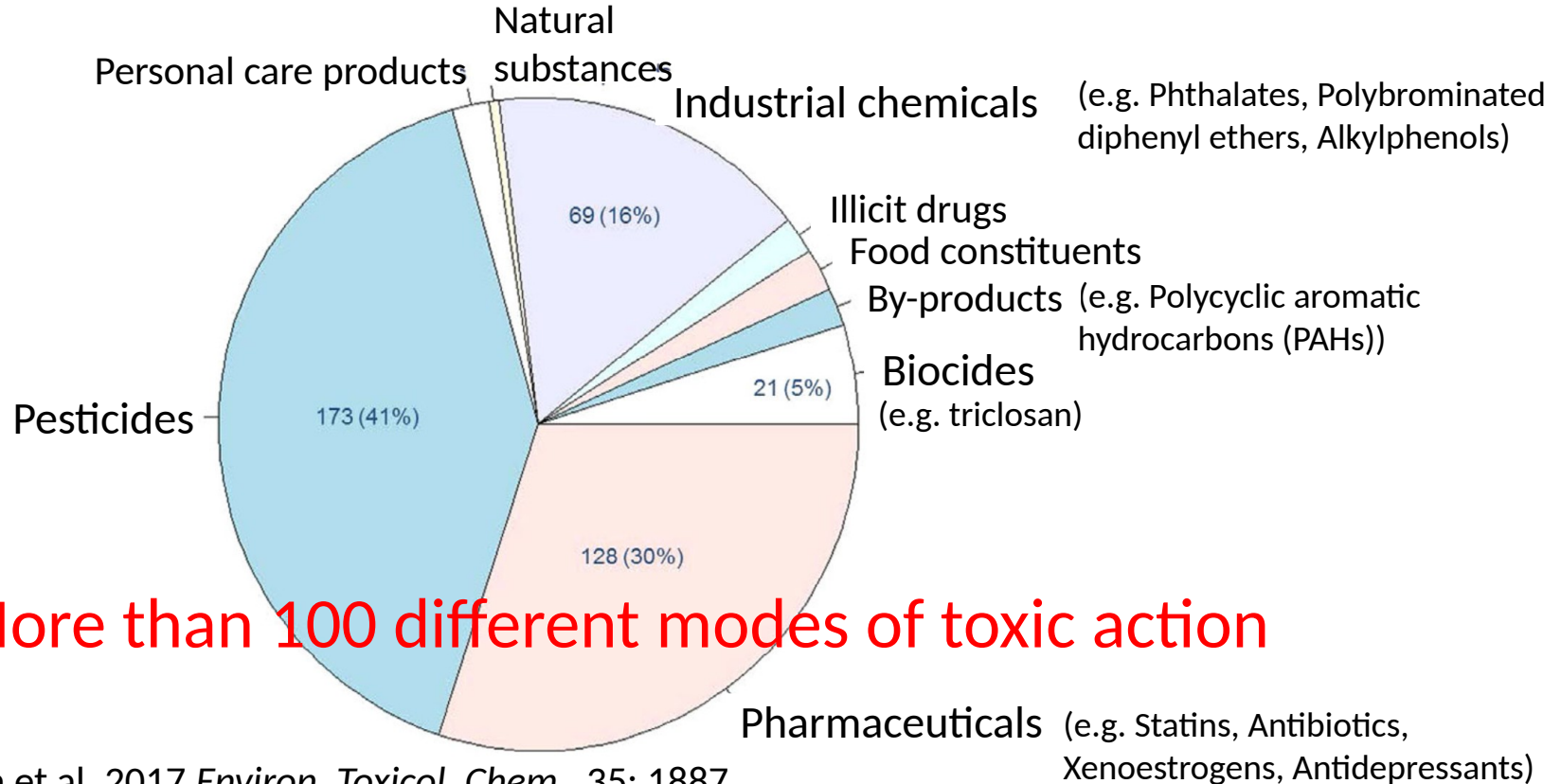
High diversity of modes of action

426 compounds found in 3 European river basins

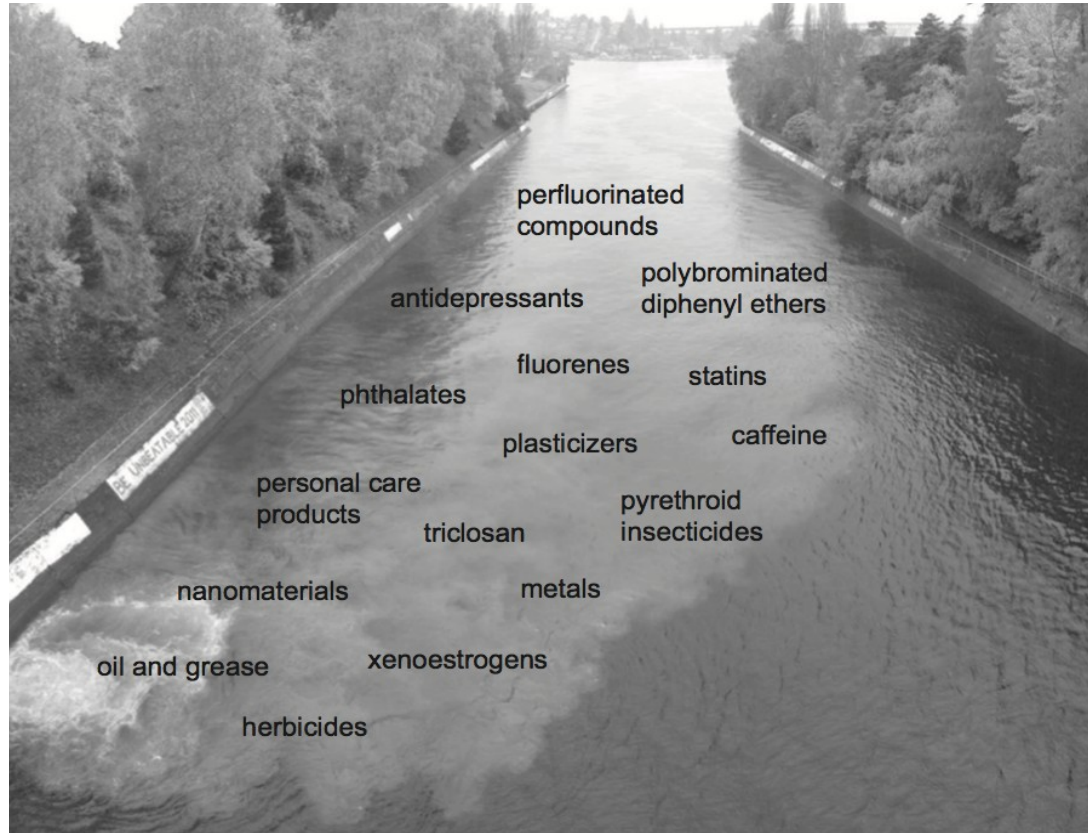


High diversity of modes of action

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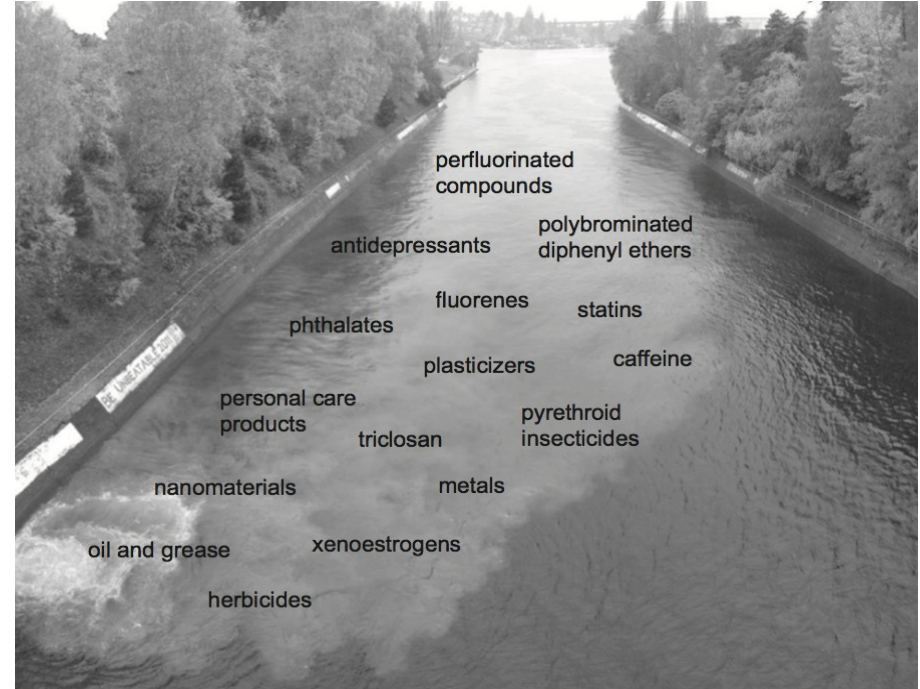
How to assess the joint effect of chemicals?



How to assess the joint effect of chemicals?

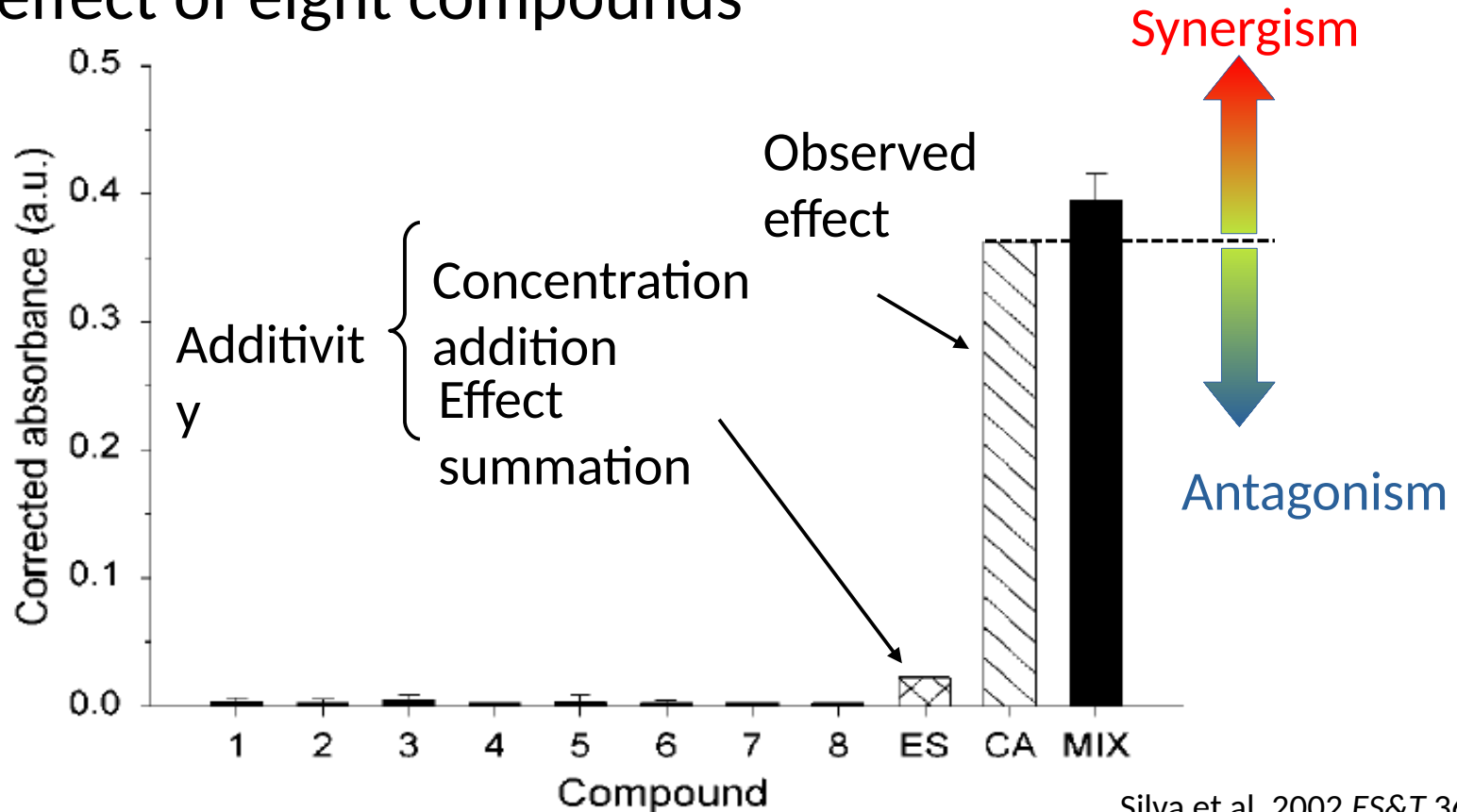
Imagine that we detected 20 chemicals at concentrations $c_1, c_2 \dots c_{20}$. What data would you need as a prerequisite to estimate their joint effect and how could you calculate a joint effect?

Discuss 5 minutes with your neighbour and then report to class.



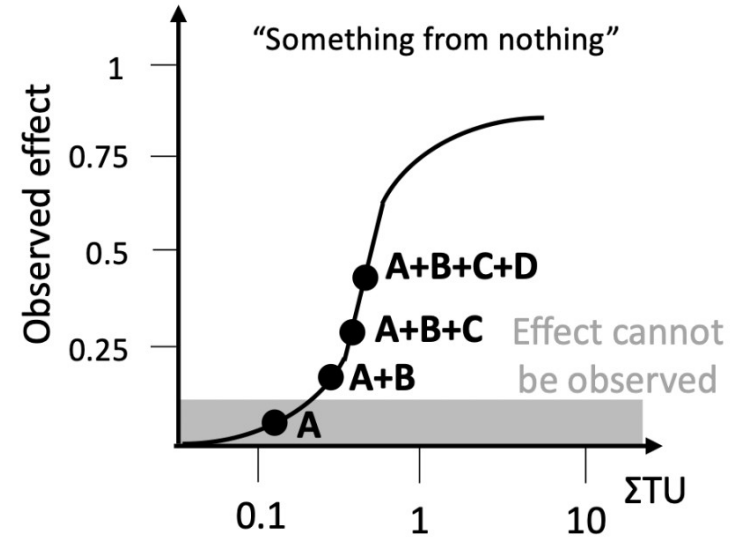
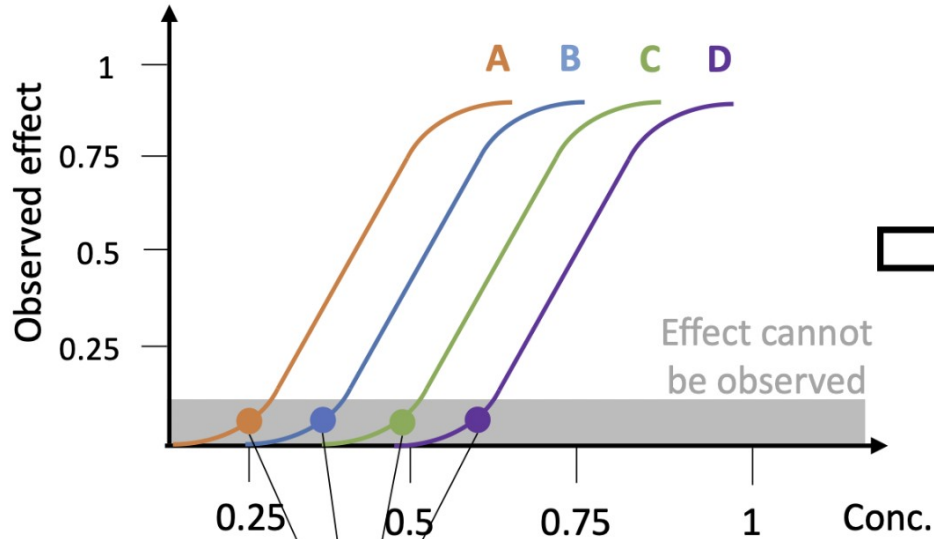
Mixture toxicity can matter!

Joint effect of eight compounds

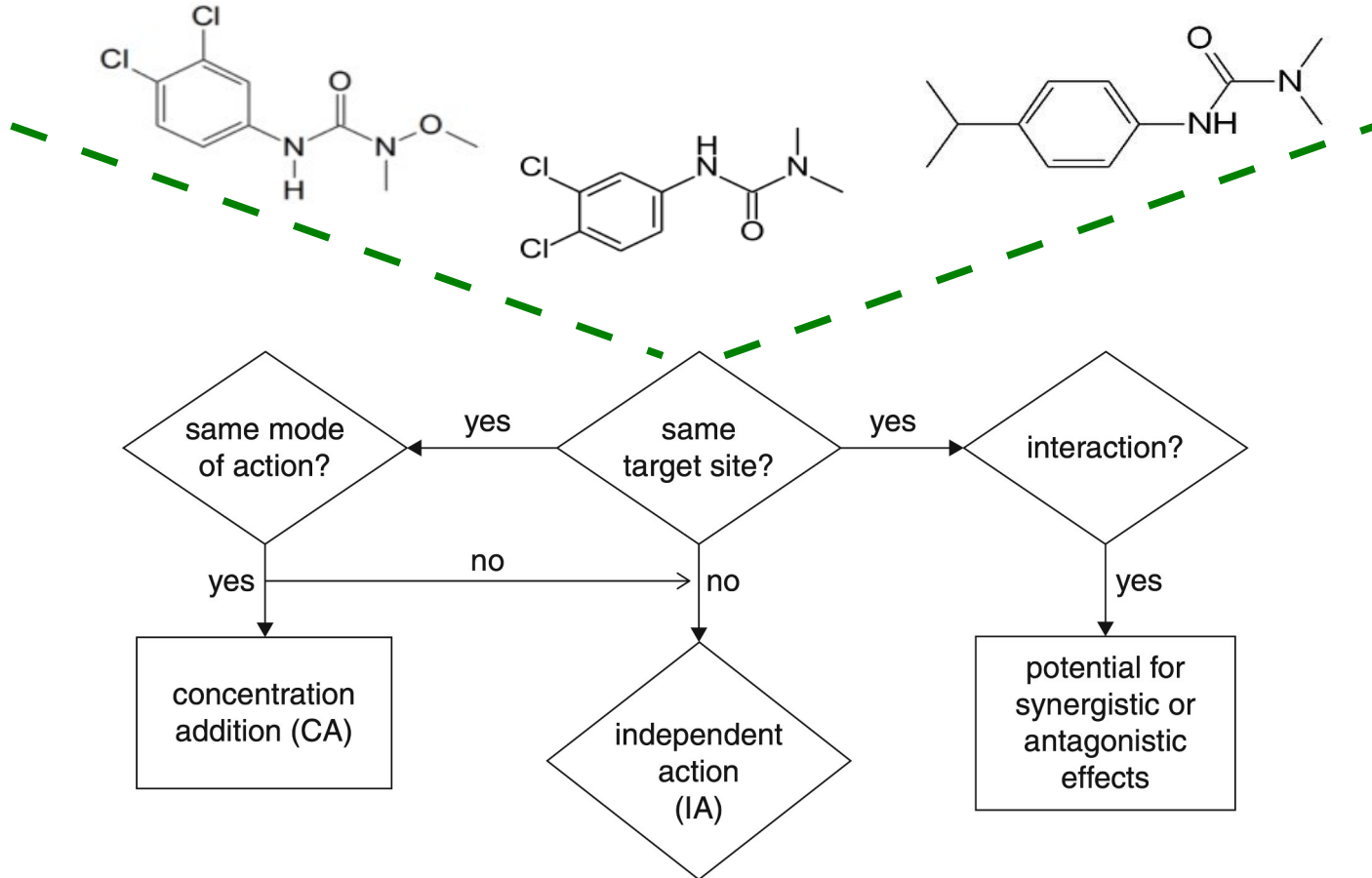


Something from nothing effect

Individual chemicals



Mixtures and mode of action



Concentration addition (CA)

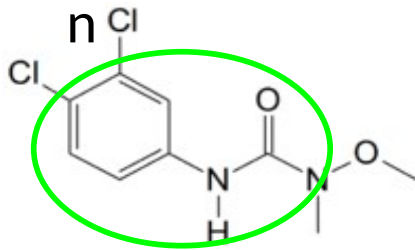
$$ECx_{mixCA} = \left(\sum_{i=1}^n \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

c_i : concentration of compound i
 c_{mix} : concentration of the mixture
 ECx_i : effect concentration x% for i
 ECx_{mixCA} : mixture concentration having x% of effects following CA

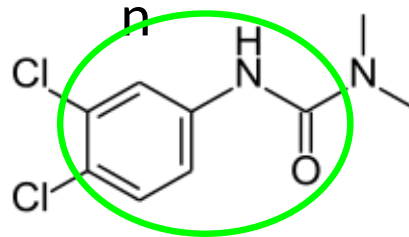
Example: Phenylurea herbicides

Chevre & Gregoria
2013

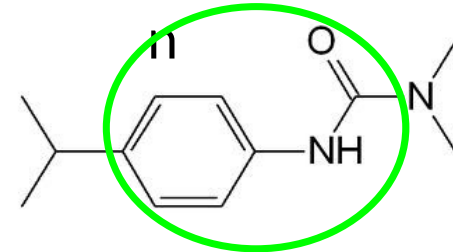
Diuro



Linuro



Isoproturo



Algae 72-h EC50,
growth ($\mu\text{g L}^{-1}$)

2.7

16

13

Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^n \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

Observed conc. ($\mu\text{g L}^{-1}$)	5.4	160	1.3
p_i	$\frac{5.4}{166.7} = 0.032$	$\frac{160}{166.7} = 0.96$	$\frac{1.3}{166.7} = 0.008$
Algae 72-h EC50, growth ($\mu\text{g L}^{-1}$)	2.7	16	13
$\frac{p_i}{EC_{xi}}$	$\frac{0.032}{2.7} = 0.012$	$\frac{0.96}{16} = 0.06$	$\frac{0.008}{13} = 0.0006$

Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^n \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

$$\frac{p_i}{EC_{xi}} \quad \frac{0.032}{2.7} = 0.012 \quad \frac{0.96}{16} = 0.06 \quad \frac{0.008}{13} = 0.0006$$

$$\sum_{i=1}^n \frac{p_i}{EC_{xi}} \quad 0.012 + 0.06 + 0.0006 = 0.0726$$

$$ECx_{mixCA} = \frac{1}{0.0726} = 13.77$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^n \frac{c_i}{EC_{xi}}$$

c_i : concentration of compound i
 EC_{xi} : effect concentration x% for compound i

Observed conc. ($\mu\text{g L}^{-1}$)	5.4	160	1.3
Algae 72-h EC50, growth ($\mu\text{g L}^{-1}$)	2.7	16	13

$$\frac{c_i}{EC_{xi}} \quad \frac{5.4}{2.7} = 2 \quad \frac{160}{16} = 10 \quad \frac{1.3}{13} = 0.1$$

$$sumTU = 2 + 10 + 0.1 = 12.1$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^n \frac{c_i}{EC_{xi}}$$

c_i : concentration of compound i
 EC_{xi} : effect concentration x% for compound i

$$sumTU = 2 + 10 + 0.1 = 12.1$$

Same as

$$\frac{\sum_{i=1}^n c_i}{ECx_{mixCA}} = \frac{166.7}{13.77} = 12.1$$

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)]$$

$E(c_{mixIA})$: predicted effect of the mixture following IA

$E_{(ci)}$: effect of compound i at concentration c

Example: Insecticide, herbicide and fungicide

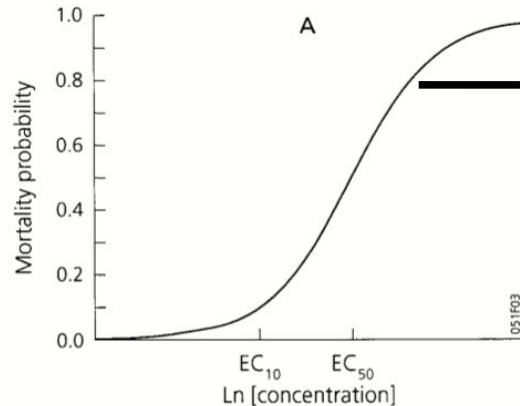
	Diuro	Thiacloprid	Tebuconazole
Observed concentrations ($\mu\text{g L}^{-1}$)	50	10	20

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)]$$

Example: Insecticide, herbicide and fungicide

	Diuro	Thiacloprid	Tebuconazole
Observed concentrations ($\mu\text{g L}$)	50	10	20



	Derive $E(c_i)$		
$E(c_i)$	0.05	0.2	0.35

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)]$$

$E(c_i)$	0.05	0.2	0.35
----------	------	-----	------

$1 - E(c_i)$	0.95	0.8	0.65
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$\prod_{i=1}^n [1 - E(c_i)]$	$0.95 * 0.8 * 0.65 = 0.49$
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$$E(c_{mixIA}) = 1 - 0.49 = 0.51$$

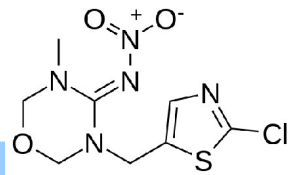
Computation of IA often hampered by data availability
(mostly only NOEC, LOEC, EC50 reported)

Exercise

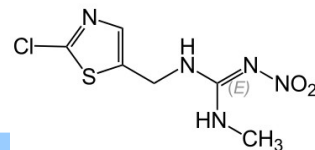
$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)] \quad sumTU = \sum_{i=1}^n \frac{c_i}{EC_{xi}}$$

Please calculate the mixture toxicity using CA as sumTU and IA models for the following situation:

Thiametoxam



Clothianidin



96-h LC50 *Chironomus riparius* (μg/L)

100

29

Observed concentrations (μg/L)

5

2.9

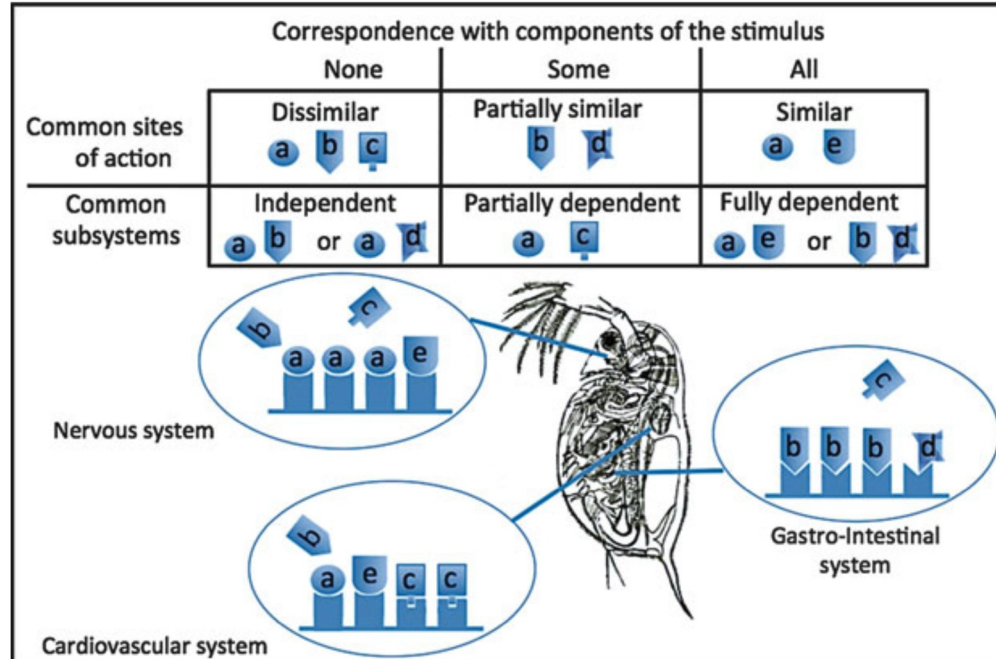
$E(c_i)$

0.02

0.2

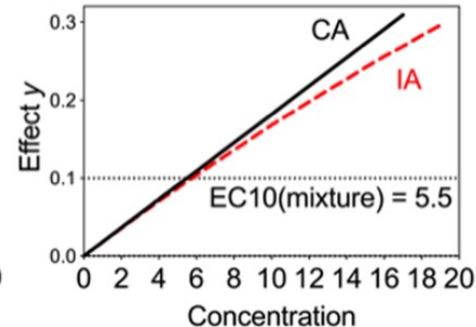
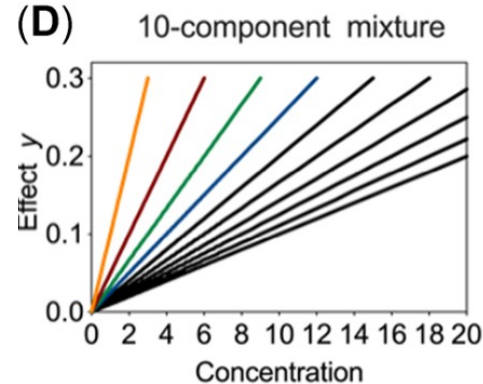
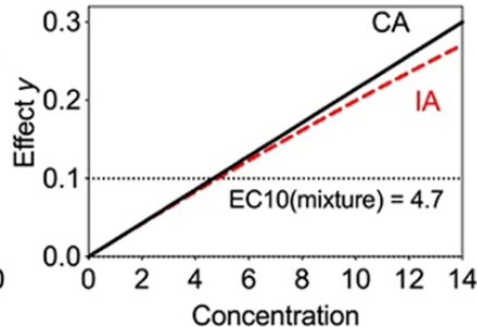
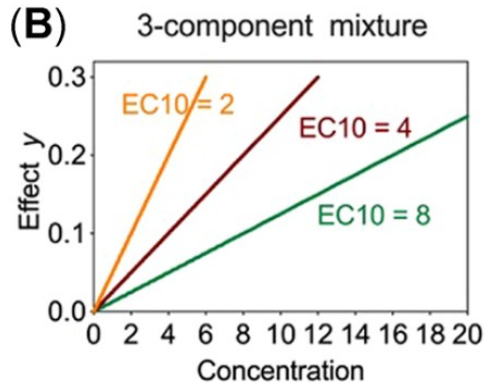
CA or IA?

- Same target site and MoA → CA
- But MoA can vary between organs



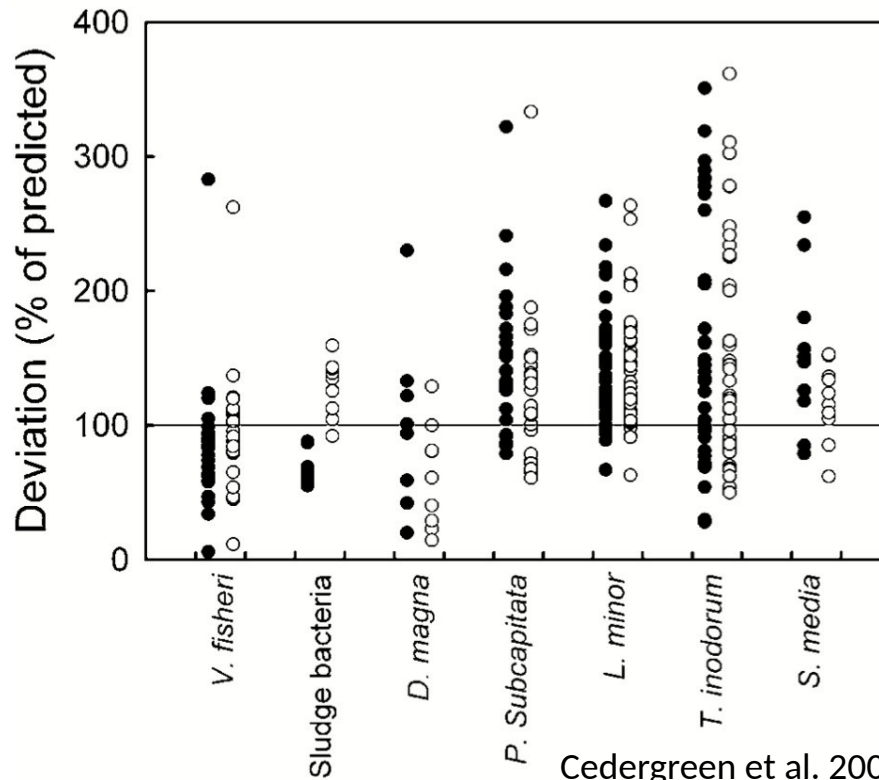
CA or IA?

- For equipotent mixtures with low effect levels (i.e. $<EC_{10}$), often predictions converge
- CA standard because data often too limited for IA



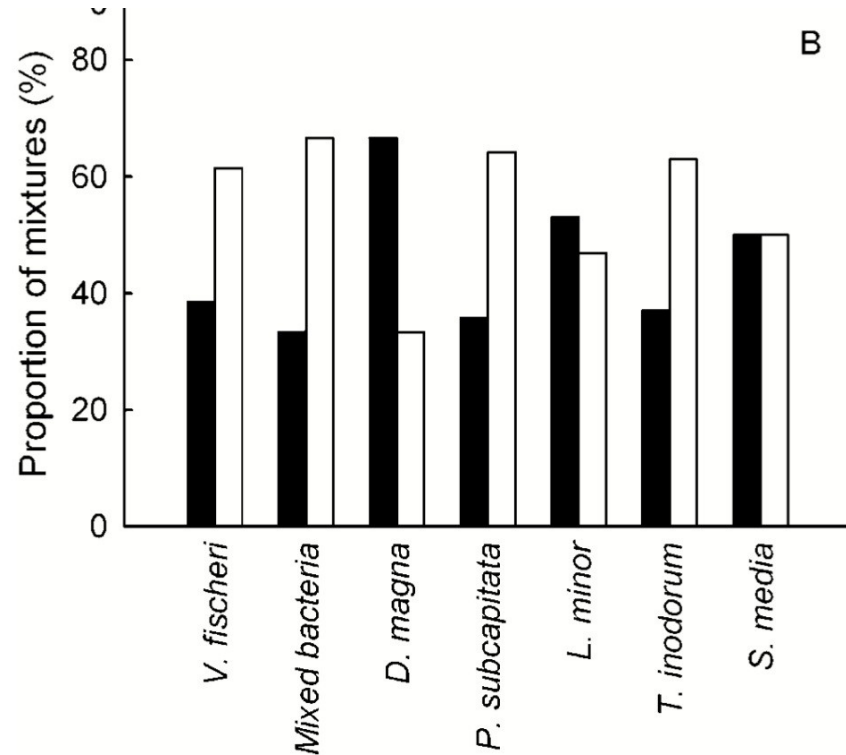
How good is the prediction?

Ratio of CA (black) and IA (white) prediction to observed EC50



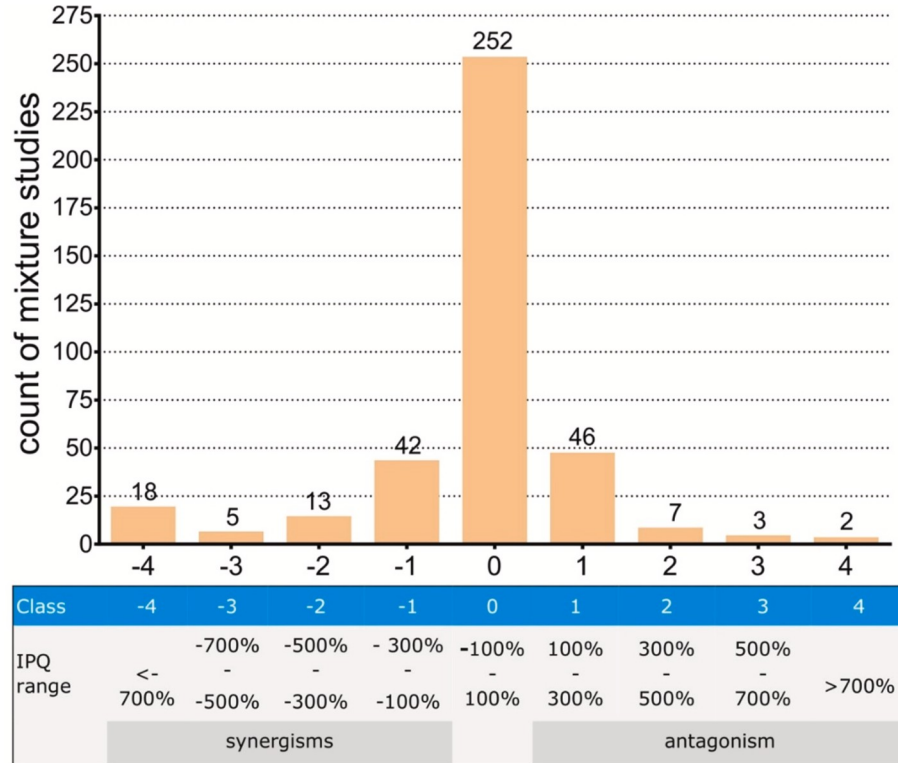
How good is the prediction?

Proportion of mixtures better described with CA (black) and better described with IA (white)



How good is the prediction?

Observations largely within 1 order of magnitude of mixture prediction



How to identify synergism and antagonism?

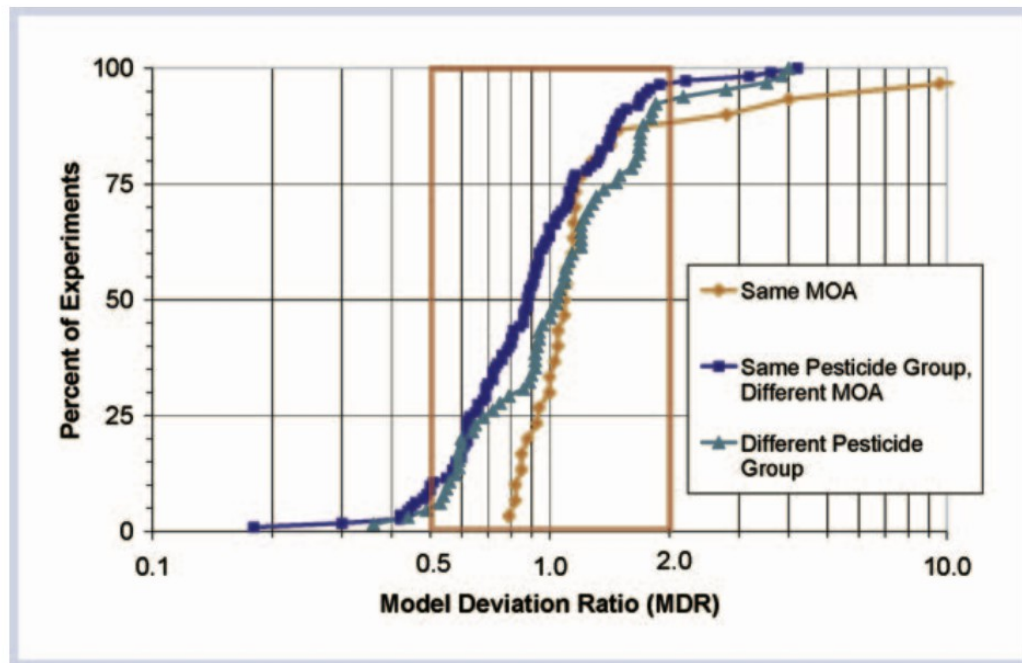
- Model Deviation Ratio (MDR) often used in Ecotoxicology:

$$\text{MDR} = \frac{\text{Expected/predicted EC with null model}}{\text{Observed EC}}$$

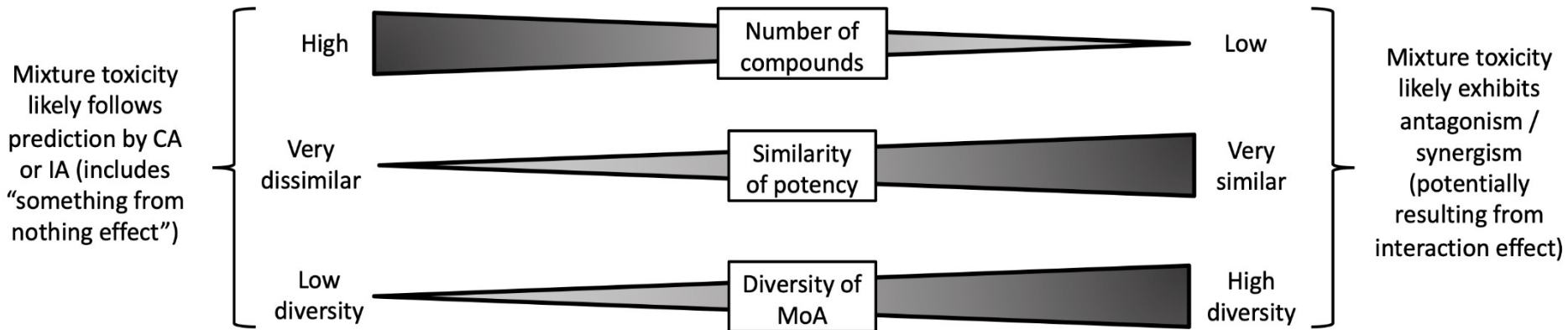
- MDR sets threshold based on effect size (e.g. 0.5 and 2)
- Why not statistical significance?
 - Sample-size dependent: e.g. low effect size (deviation from model) significant for large sample size, high effect size not significant for small sample size

How good is the prediction?

MDR for CA for pesticide mixtures → Reasonable match for majority of experiments, low fraction of synergism/antagonism



How frequent are synergism and antagonism?

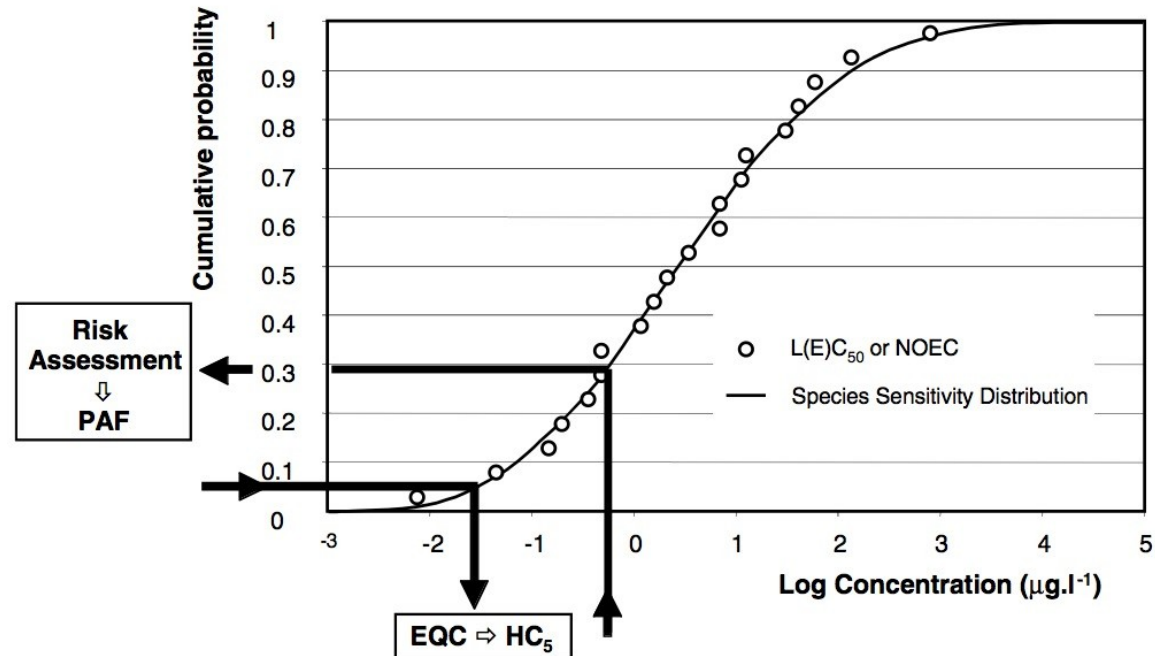


msPAF: Mixture prediction for SSDs

- Mixture prediction (CA and IA/RA) for community level effects
- Instead of ECx and $E(c)$ values, use Hcp and PAF values from single-substance SSDs

SSDs for risk assessment

- Hazardous concentration (HC_p): conc. at which $p\%$ affected
- Potentially affected fraction PAF: given by concentration



Cutoff for community PNEC

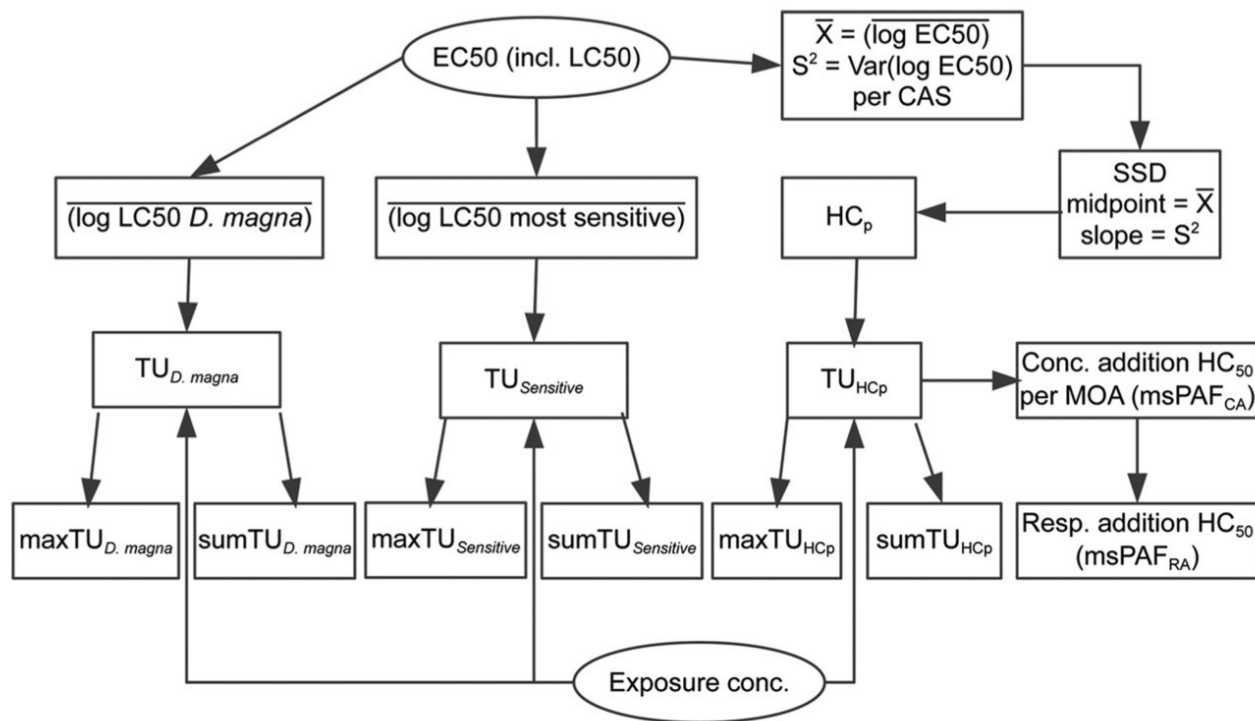
msPAF: Mixture prediction for SSDs

- Mixture prediction (CA and IA/RA) for community level effects
- Instead of ECx and $E(c)$ values, use Hcp and PAF values from single-substance SSDs
- $msPAF_{CA}$: Aggregate Hcp for all compounds with same MoA
- Alternative: $sumTU_{HCp}$

$$msPAF_{RA} = 1 - \prod_{i=1}^n (1 - msPAF_{CA,i})$$

Example for community-level mixture indices

IA/RA was only calculated in SSD context due to lack of data



Example for community-level mixture indices

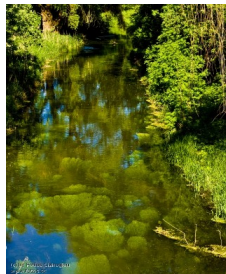
For 4 of 5 data sets $sumTU_{HC5}$ best metric, but maxTU and metrics based on *D. magna* or most sensitive test species not much worse

Exposure metric	Rank sum of metric across countries	Number of times metric was among models within a range of 2 to BIC of best-fit model
$sumTU_{HC5}$	14	4
$sumTU_{Sensitive}$	18	3
$maxTU_{HC5}$	21	3
$maxTU_{Sensitive}$	21	3
$maxTU_{D.magna}$	22	2
$sumTU_{D.magna}$	26	3
$msPAF_{RA}$	32	1
$maxTU_{HC50}$	35	1
$sumTU_{HC50}$	36	1

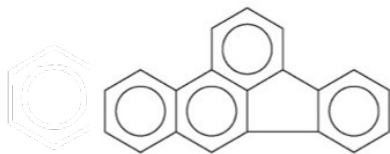


Chemicals don't occur in isolation

Analysis of german WFD monitoring data



Excessive
nutrients



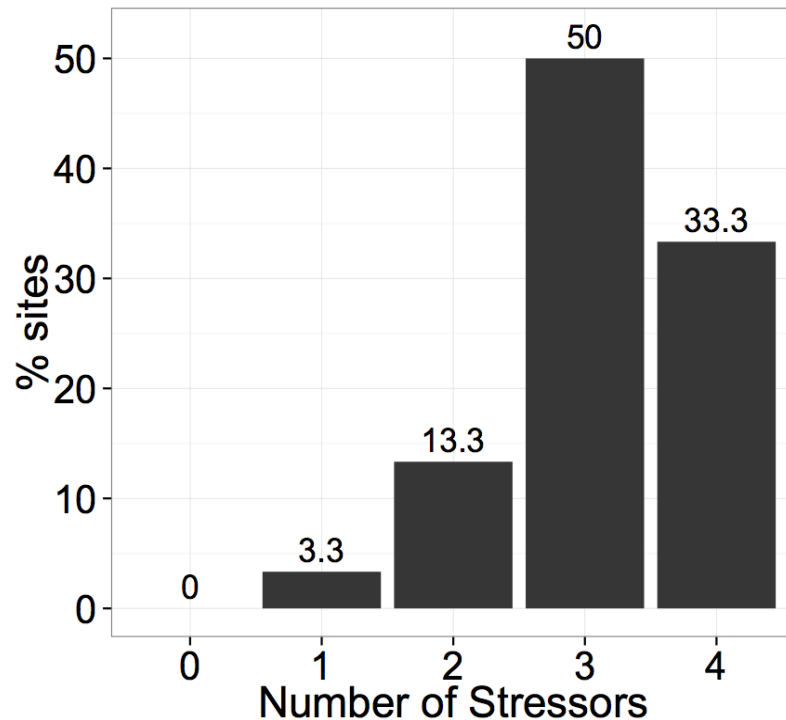
Toxicants



Invasive
species

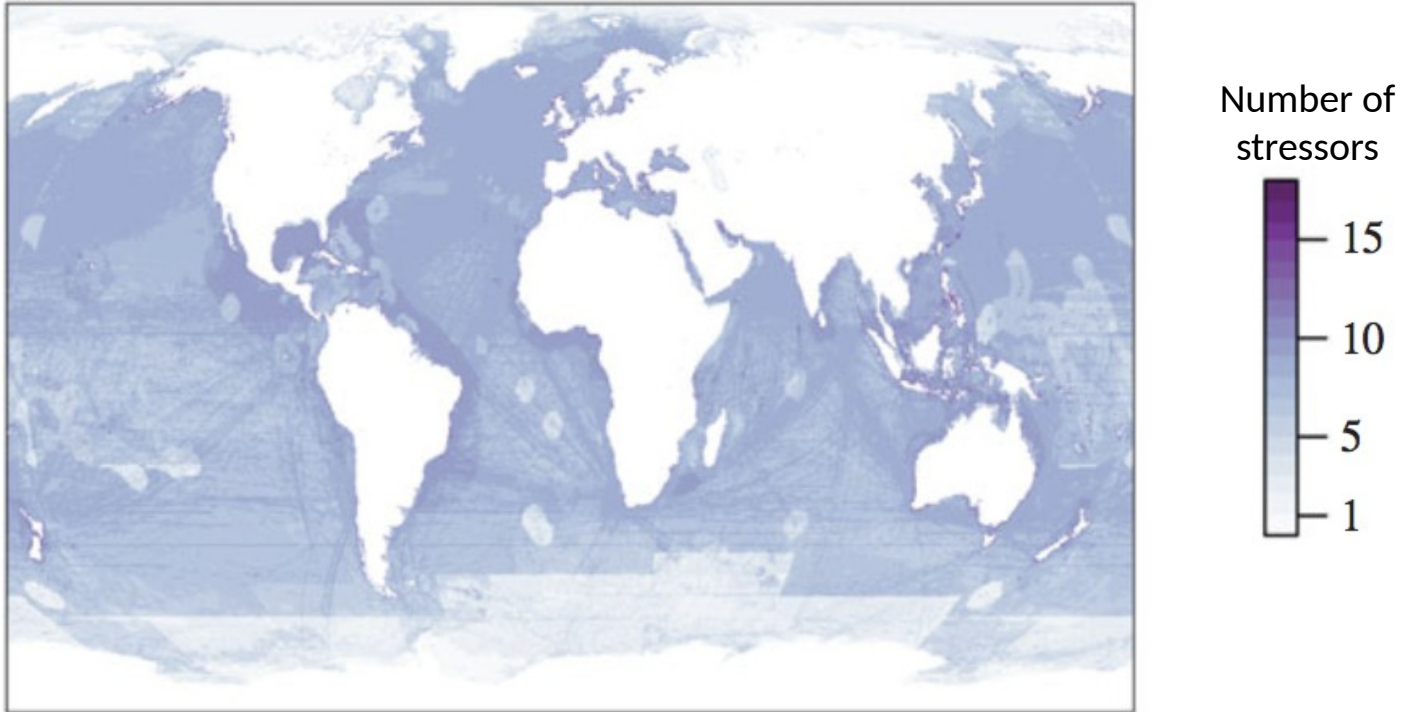


Habitat
degradation



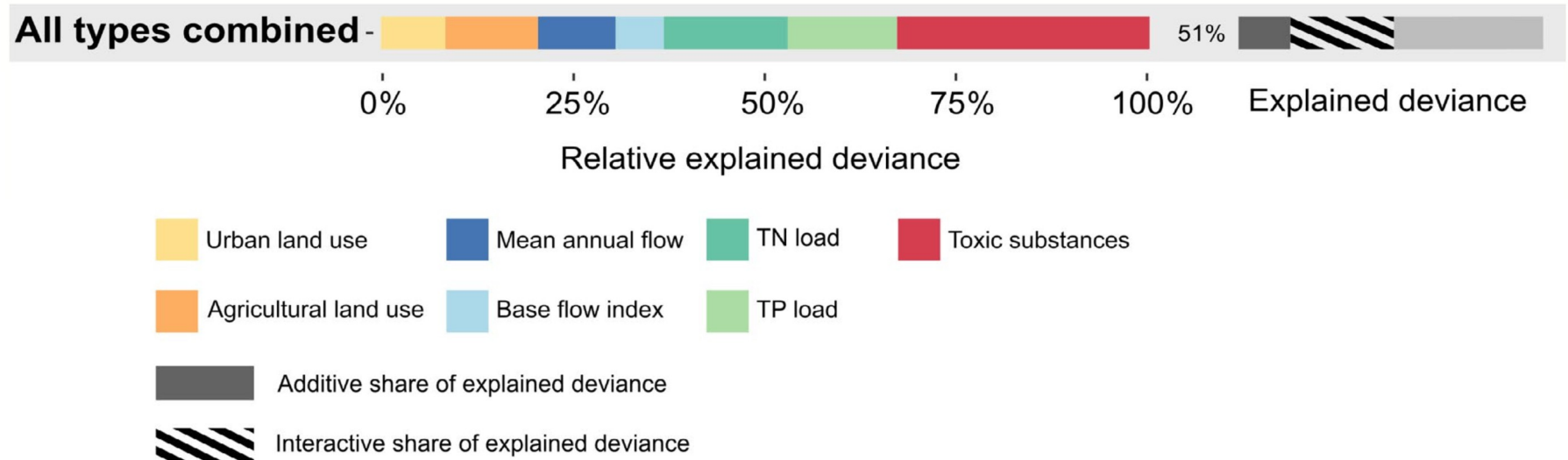
Chemicals don't occur in isolation

(a) Global distribution of marine stressors



Chemicals don't occur in isolation

- Analysis of European WFD monitoring data
- Several stressors/drivers of stressors explain variability in ecological status including mixtures
- Relevant fraction of interactions



A wide range of chemical mixtures
with different MOAs and multiple
non-chemical stressors co-occur
that can potentially affect
organisms and may interact

What are mechanisms underlying interactions?

Two stressors or chemicals can interact directly or indirectly regarding their (direct) effect

Direct interaction:

Stressor/Chemical *A* moderates intensity of Stressor/Chemical *B*

What are mechanisms underlying interactions?

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Direct interaction:

Stressor/Chemical A moderates intensity of Stressor/Chemical B

Examples

+ Sand



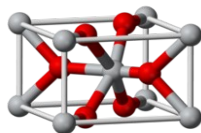
Reduces

Predation by



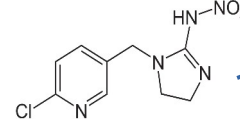
Louhi et al. 2017 *Can. J. Fish. Aqu. Sci.* 74: 32

TiO₂ NP



Reduces

Concentration of



antagonism



What are mechanisms underlying interactions?

Two stressors or chemicals can interact directly or indirectly regarding their (direct) effect

Indirect interaction:

Stressor/Chemical *A* moderates sensitivity to Stressor/Chemical *B*

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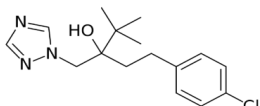
Examples



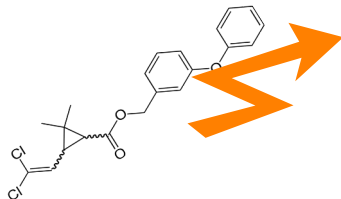
Reduces capacity
to deal with



Orcutt & Porter 1984 *Oecologia* 63: 300



Inhibits
biotransformation of



Cedergreen et al. 2017 *ES&T* 51:14379



synergism



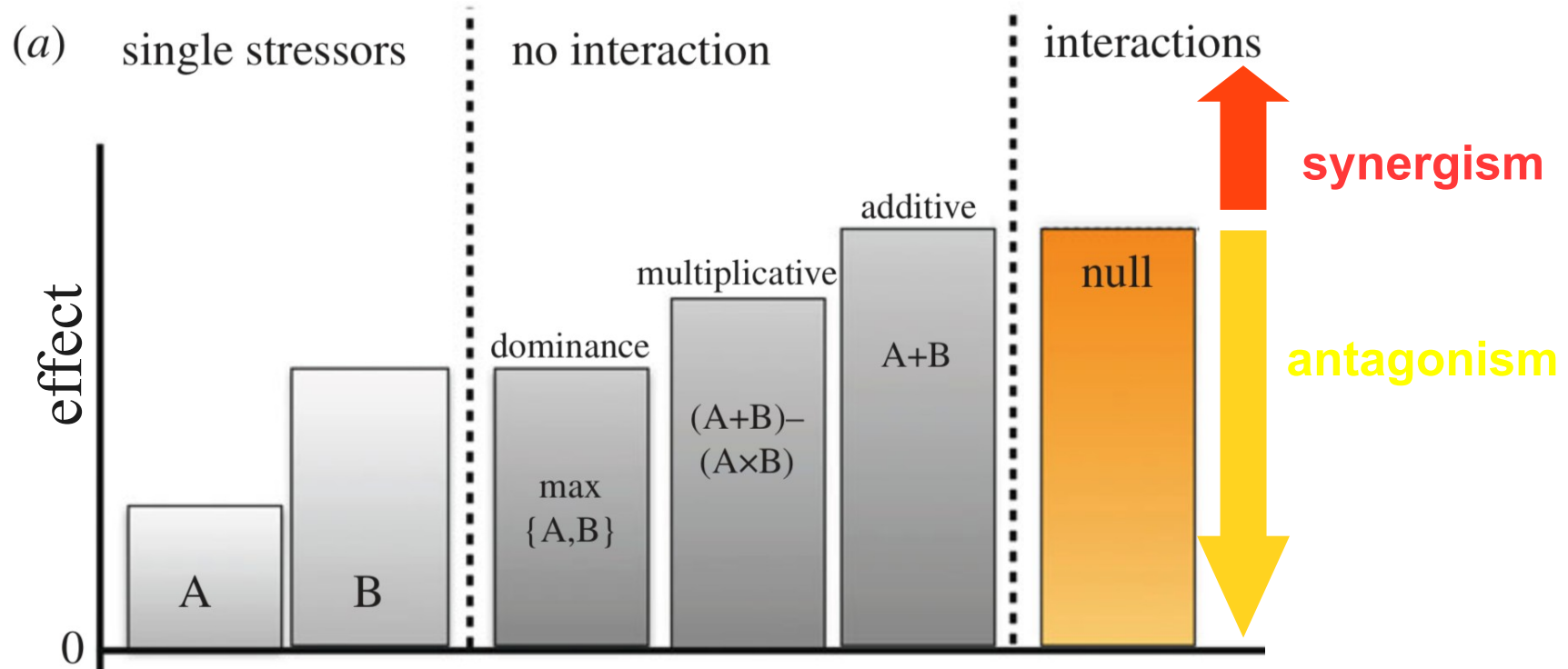
How frequent are synergism and antagonism?

Meta-analysis of 15 meta-analyses on multiple stressors:

- between 4% and 68% synergistic responses
- few consistent responses (no antagonism on physiological level, no synergism on community- or ecosystem level)

Synergism and antagonism in multiple stressor research

Interaction = (statistically significant) deviation from null model

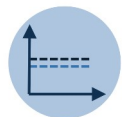


Joint challenge, different approaches

- Multiple stressor/ ecological and ecotoxicological null models have different null model standards
- Novel null model developed to integrate chemical and non-chemical stressors

Ecological model	(Eco)toxicological model	Null model equation for multiple stressor research	Required input data
Simple addition	–	$f_{AB}(SI_A, SI_B) = f_A(SI_A) + f_B(SI_B)$	Effects for A and B
Multiplicative	Effect addition	$f_{AB}(SI_A, SI_B) = f_A(SI_A) + f_B(SI_B) - f_A(SI_A) f_B(SI_B)$	Effects for A and B as proportions (%) ^b
Dominance	–	$f_{AB}(SI_A, SI_B) = \max(f_A(SI_A), f_B(SI_B))$	Effects for A and B
–	Concentration addition ^a (CA)	$f_{AB}(SI_A, SI_B) = f_A(SI_A + \gamma SI_B)$ with $\gamma = SI_{x_A}/SI_{x_B}$	Stressor–effect relationship for A, stressor intensity for A and B and stressor intensities SI_{x_A} and SI_{x_B} that result in the same effect size x ($x = f_A(SI_{x_A}) = f_B(SI_{x_B})$), where x is typically selected as half the effect limit
–	Stressor addition (SAM)	$f_{AB}(SI_A, SI_B) = F_{Strcap}(F_{Strcap}^{-1}(f_A(SI_A)) + F_{Strcap}^{-1}(f_B(SI_B)))$	Generalized density function for the distribution of the stress capacity f_{Strcap}^c and effects for A and B ($f_A(SI_A)$ and $f_B(SI_B)$)

Joint challenge, different approaches



Null models



Multiple stressor research

tools and methods

- ◆ Significance testing
- ◆ Null models integrating communities developed

Joint challenges & research gaps

Too focused on interactions

Null model selection



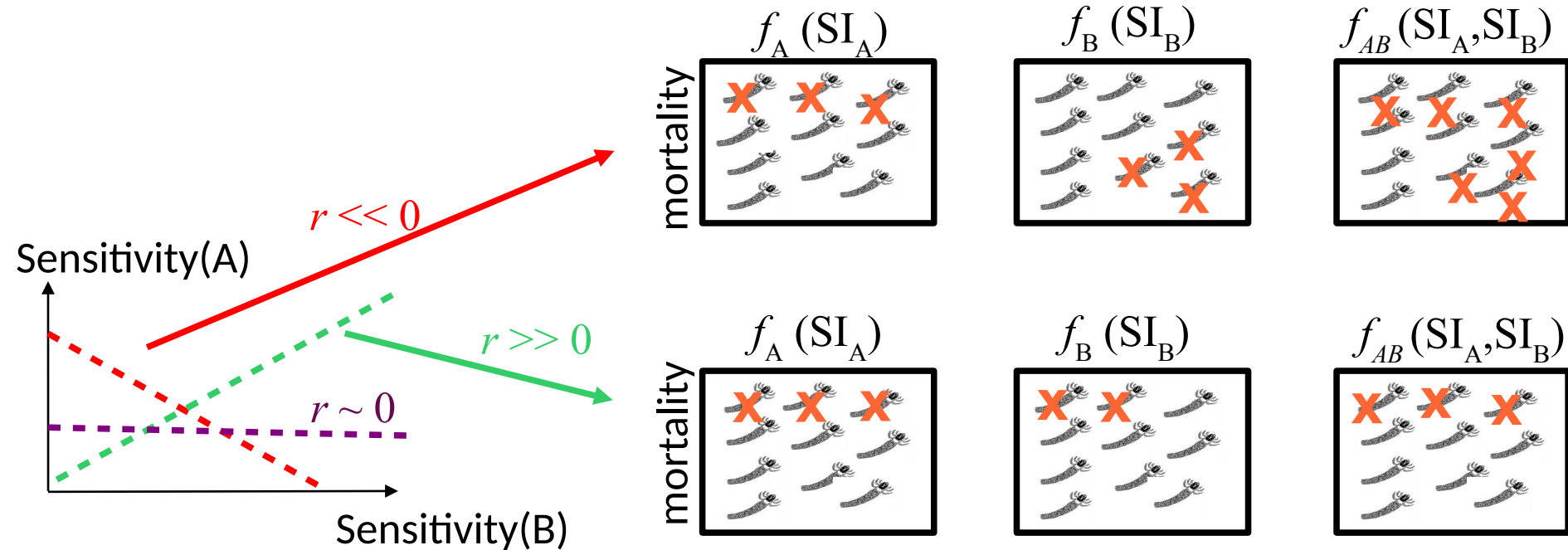
Chemical mixture research

tools and methods

- ◆ Concept of Model Deviation Ratio
- ◆ Null models integrating chemical & non-chemical stressors
- ◆ Large databases on chemical effects

How to choose a null model?

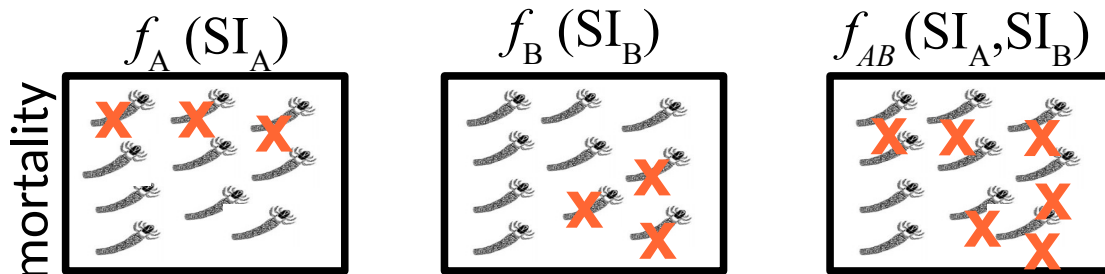
Correlations among stressor sensitivities may help



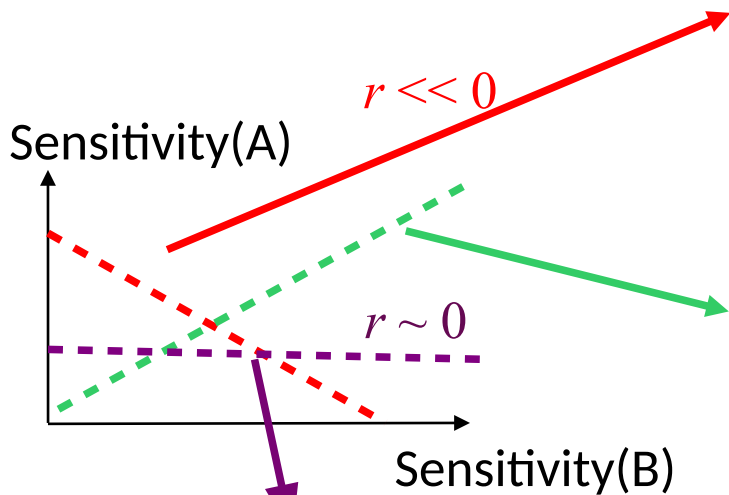
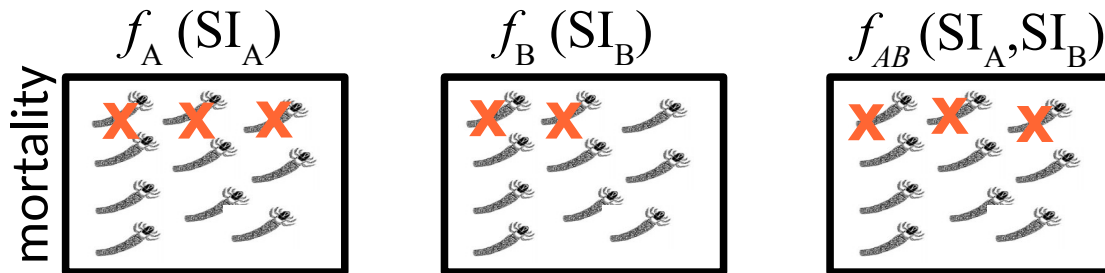
How to choose a null model?

Correlations among stressor sensitivities may help

Additive null model



Dominance null model



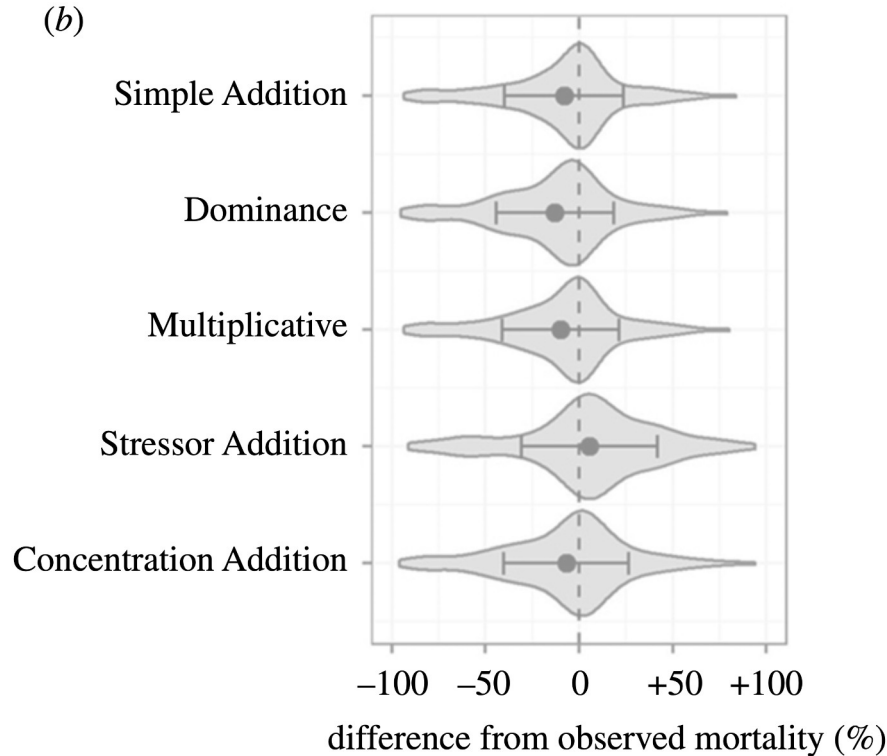
Multiplicative null model

How to choose a null model?

- Stressor mode of action: How do the stressors affect the organisms? (e.g. physically, specific or non-specific physiologically) → Similar or different mode of action? Stressor interactions known?
- Sensitivities: Relationship between sensitivities to stressors?
- Effect type: What is affected? (e.g. mortality rather multiplicative, growth rather additive)
- Effect size: What are the individual effect sizes? (e.g. close to limit or differs by orders of magnitude → dominance)
- Knowledge of stressor-effect relationship: Broader range of models available

Null model comparison

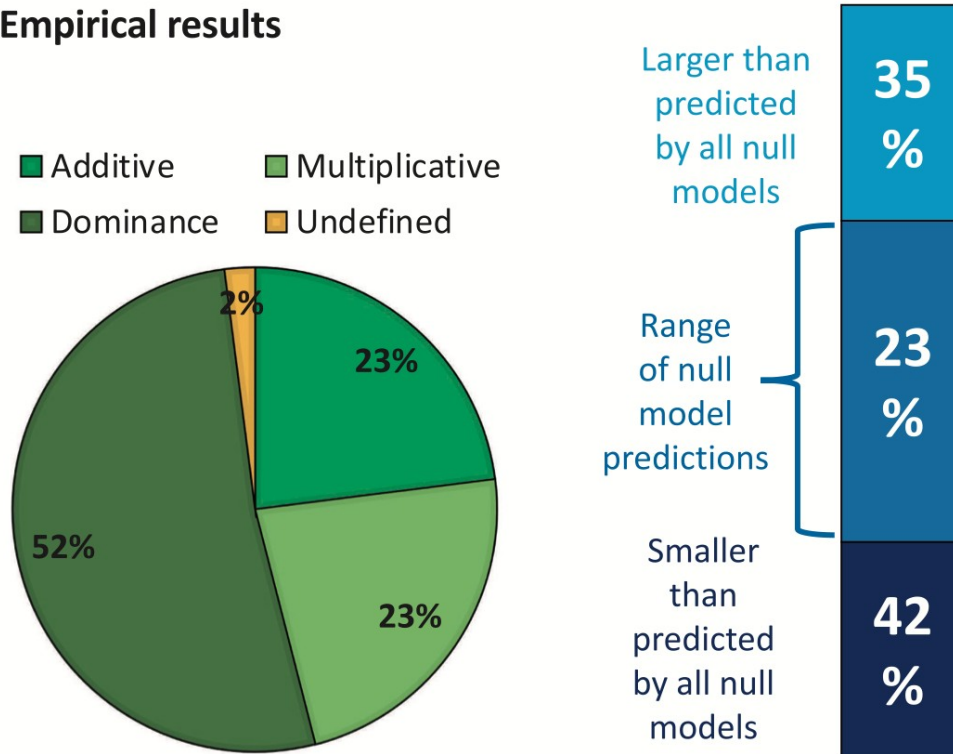
SAM (and Concentration Addition model) least biased in meta-analysis



Null model comparison

Dominance best for warming and 2nd stressor (incl. contamination)

(d) Empirical results



How frequent are synergism and antagonism?

Quite frequent even if considering range of null models

Meta-analysis of
studies on warming
and 2nd stressor

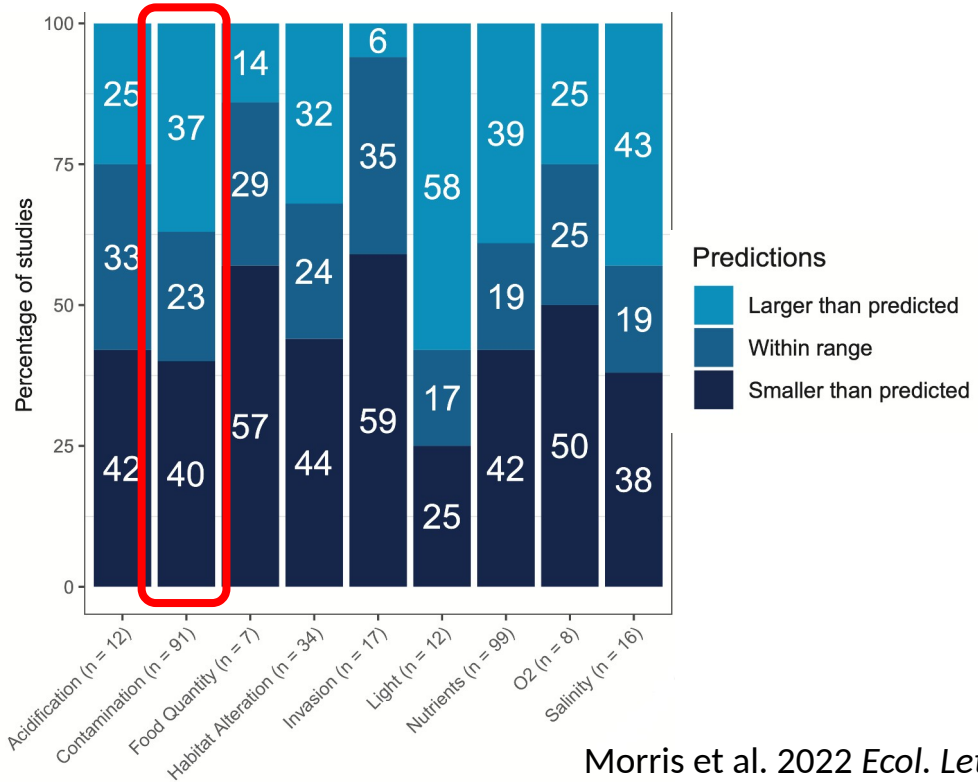




Illustration of null models

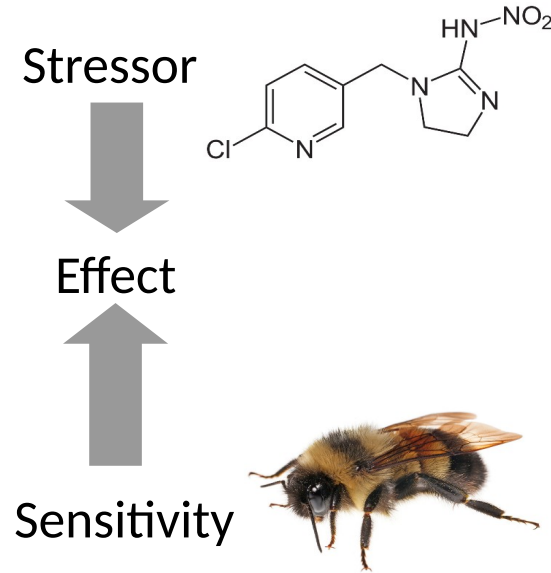


Illustration of null models: Standard models

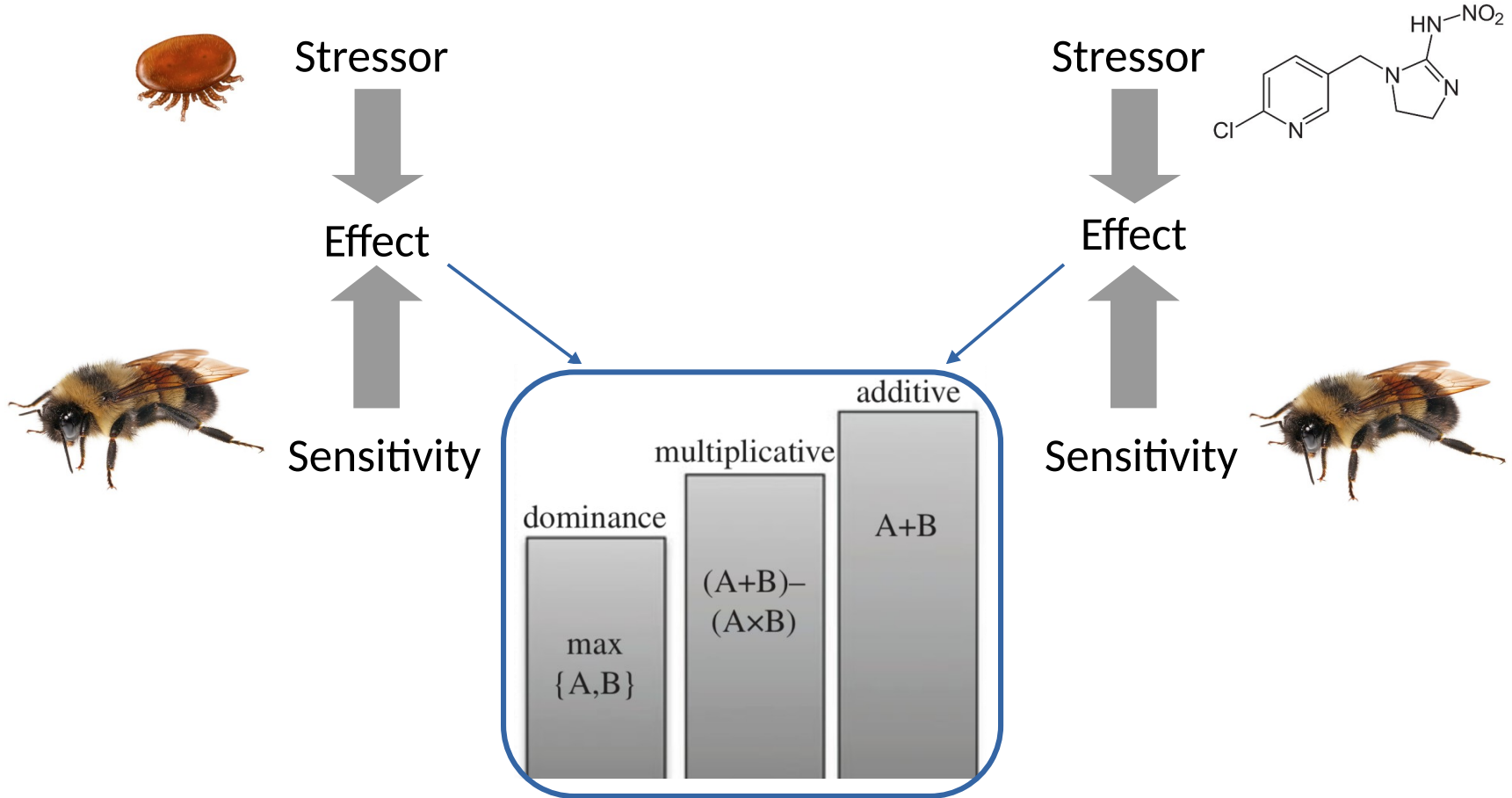
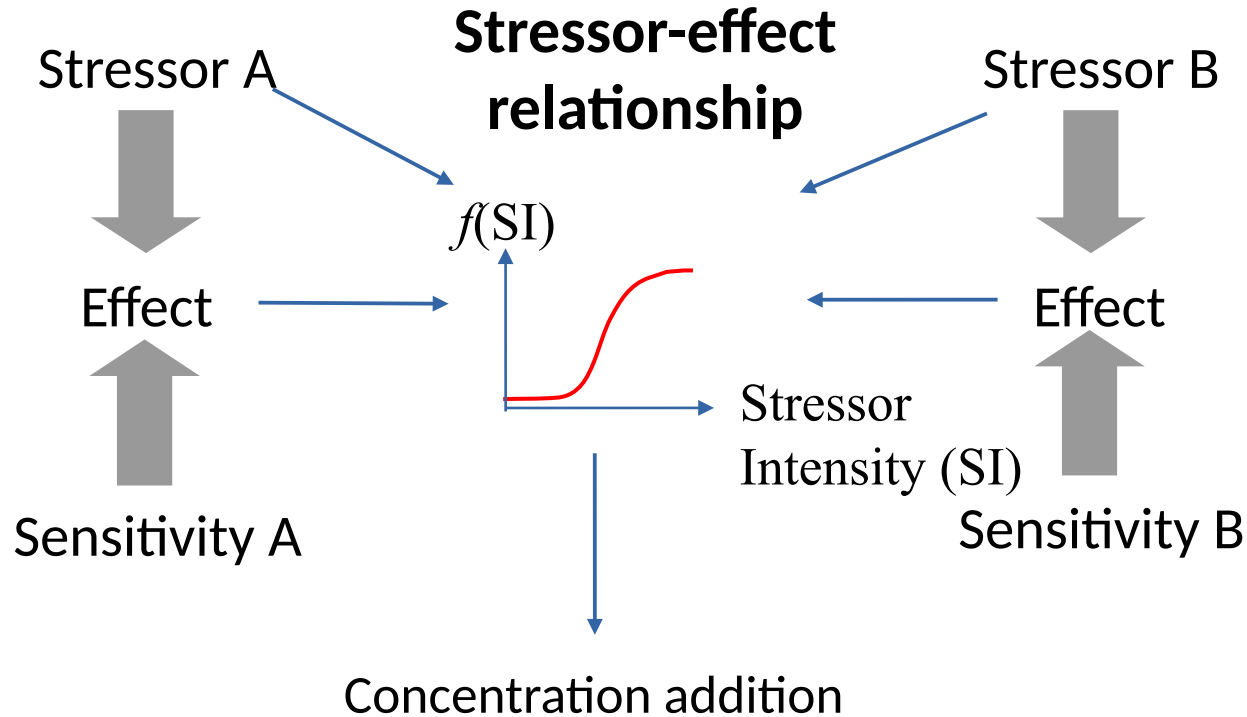


Illustration of null models: Concentration addition



Concentration addition: Adding up stressor intensities

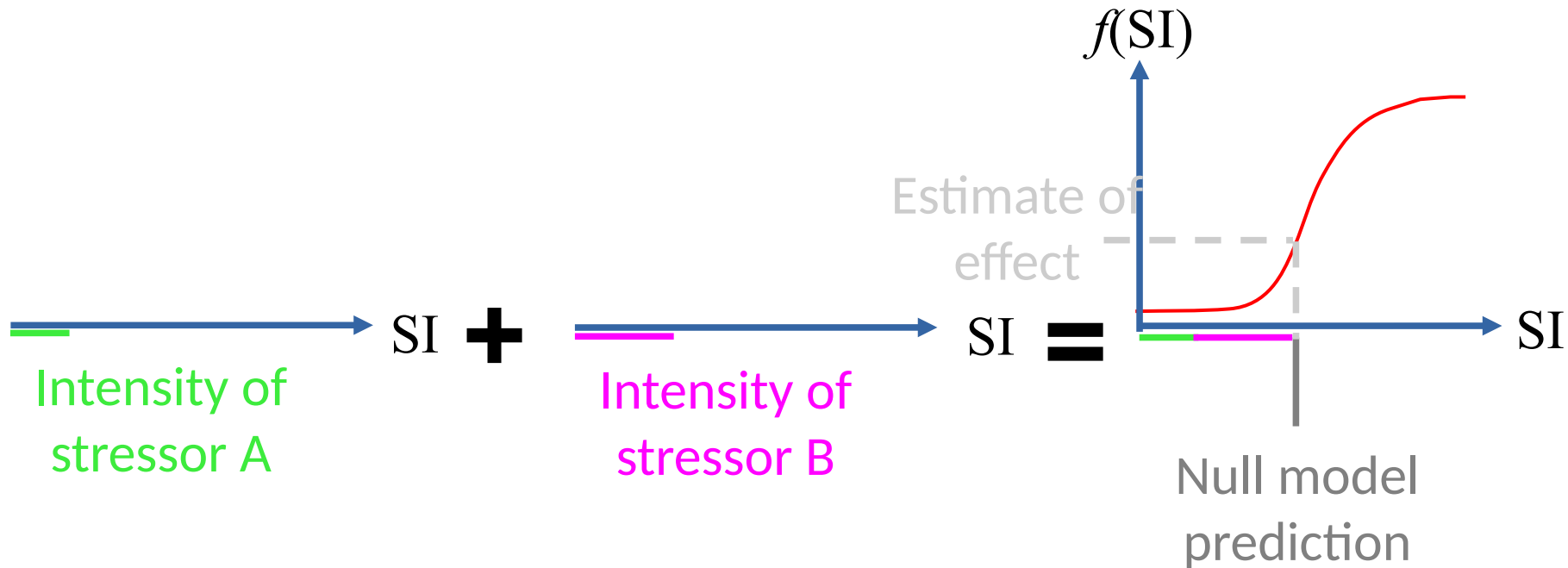
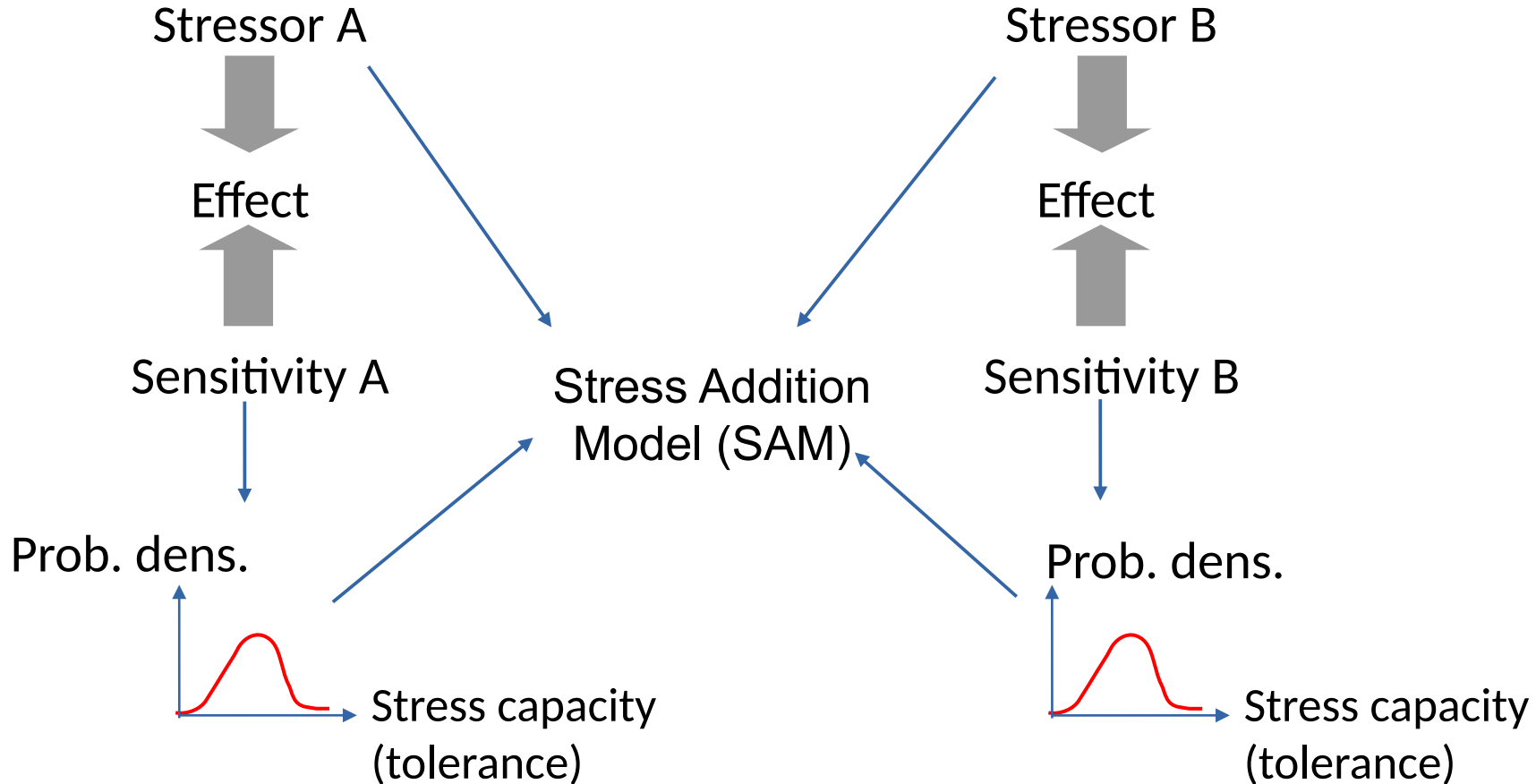
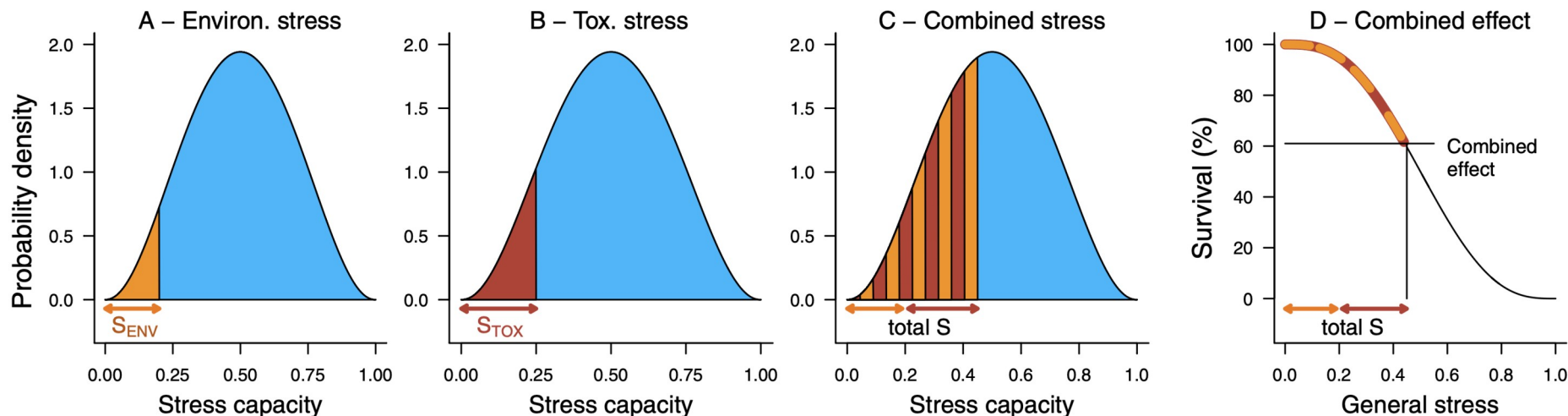


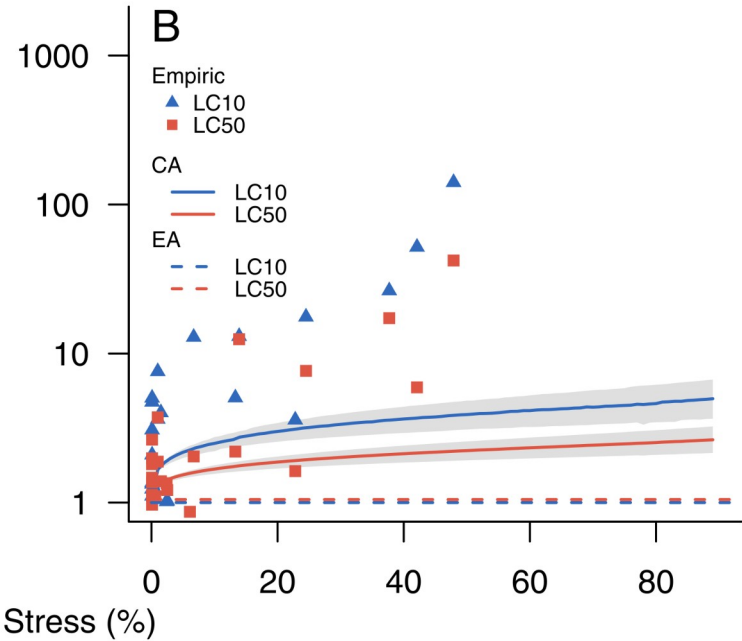
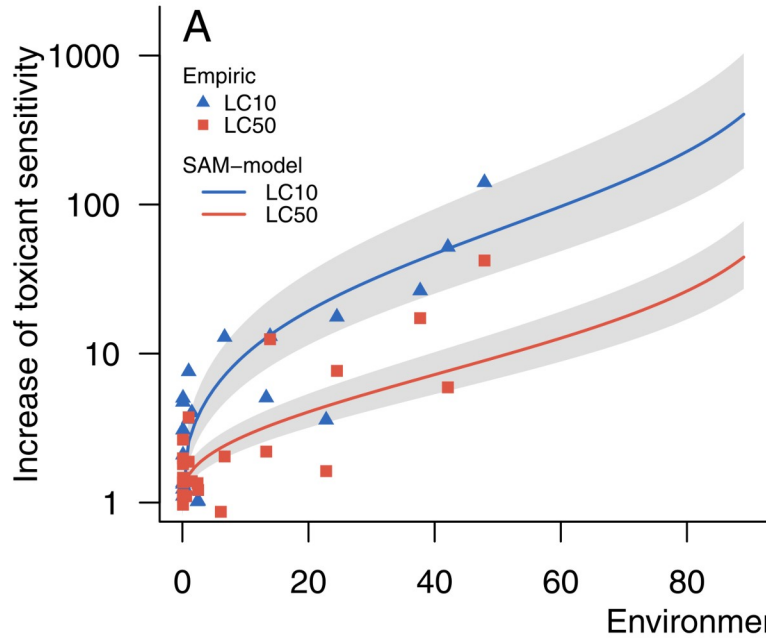
Illustration of null models: Stressor addition



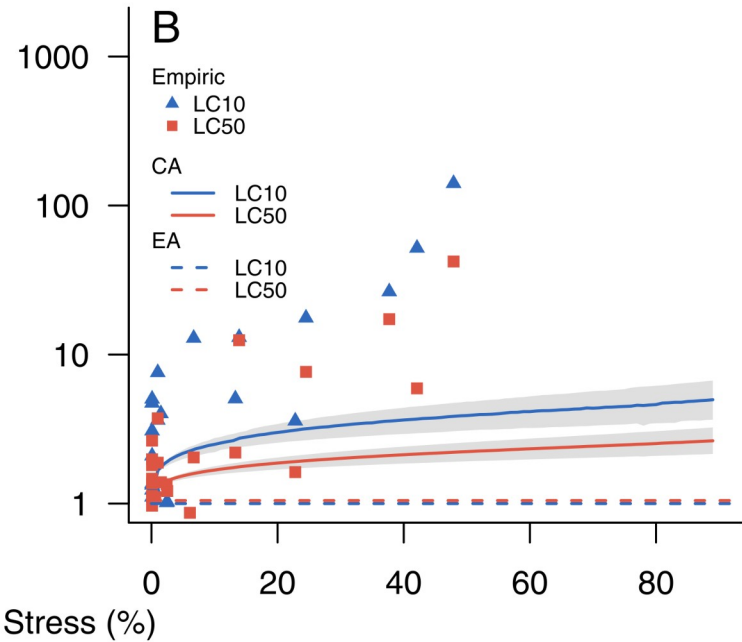
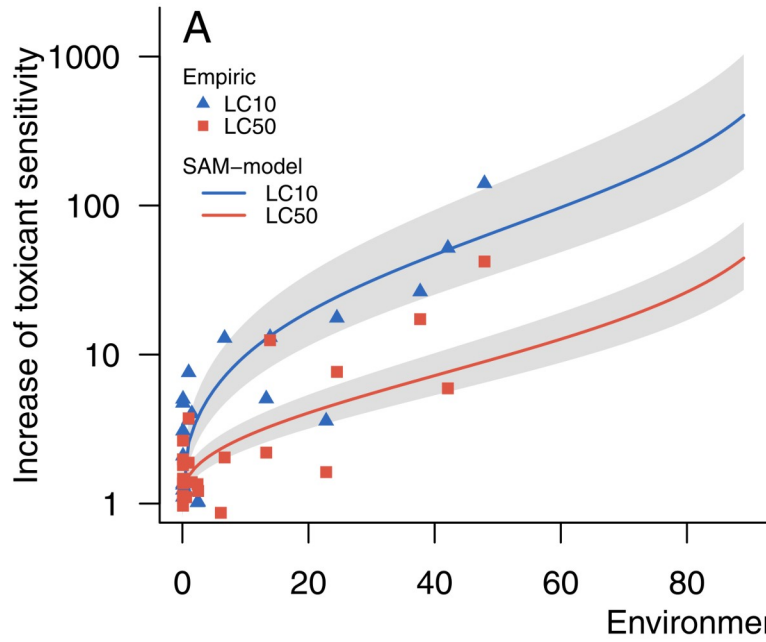
Stressor addition: Adding up stressor capacities



Stressor addition: Adding up stressor capacities



Stressor addition: Adding up stressor capacities



SAM and CA rarely considered in multiple stressor/ ecological studies, because large majority of studies too few treatment levels
→ Use regression design

Griffin et al. 2016 *Mar. Ecol. Progr. Ser.* 543: 273,
Orr et al. 2024 *Ecol. Lett.* 27: e14463

Liess et al. 2016 *Sci. Rep.* 6, 32965

Theoretical Ecotoxicology

Session 5: Mixture and multiple stressor modelling

University Duisburg-Essen SS 2025

Ralf B. Schäfer

Ecotoxicology, Faculty of Biology, UDE & RC One Health Ruhr

Intended learning outcomes of today's lecture

- Understanding and applying the major concepts mixture toxicity estimation
- Knowledge on the predictive power of mixture models
- Understanding the conceptual basis of multiple stress effects and null models

Learning targets and study questions

- Understanding and applying the major concepts mixture toxicity
 - What is the *something from nothing* effect?
 - Explain the concepts of IA, CA and sum TU.
 - Discuss the choice between IA and CA.
 - What are antagonism and synergism?
- Knowledge on the predictive power of mixture models
 - Outline shortcomings of CA and IA.
 - How well do CA and IA match with observations, and how strong is the deviation?

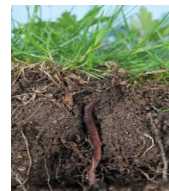
Learning targets and study questions

- Understanding the conceptual basis of multiple stress effects and of null models
 - Explain why the choice of the null model matters for the evaluation of synergism and antagonism.
 - Describe some commonly used null models.
 - Which criteria can support the choice of the null model?

Mixtures abundant in ecosystems

76 of 82 target chemicals found in soils around Paris

Gaspéri et al. 2022 *Environ. Sci. Poll. Res.* 25: 23559



www.fao.org



398 of 2316 target chemicals found in Greek waste water samples

Gago-Ferrero et al. 2020 *J. Haz. Mat.* 387: 121712

www.wikipedia.org

47 of 92 target pesticides found on insects in Malaise traps in German conservation areas

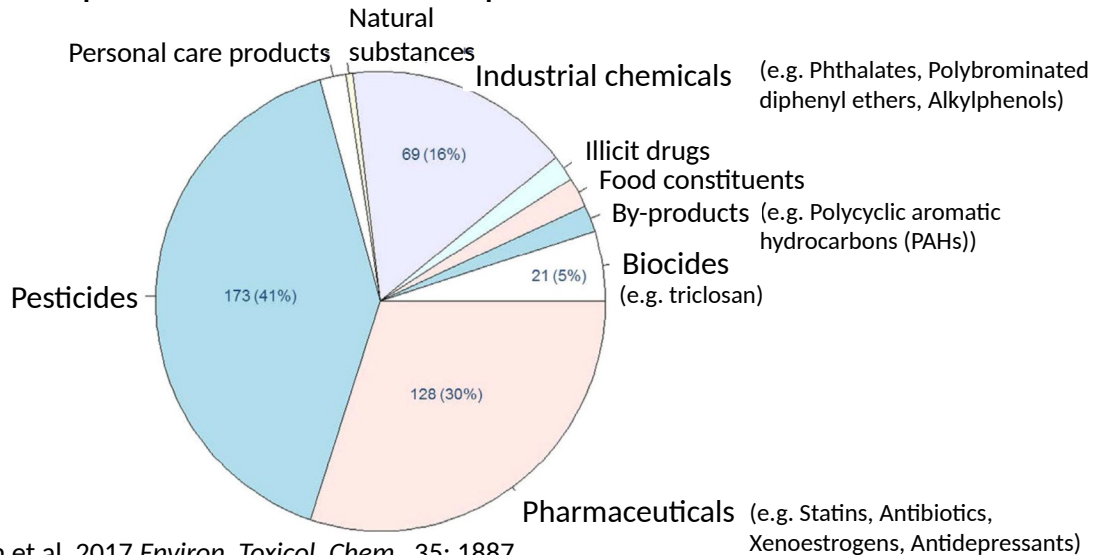
Brühl et al. 2021 *Sci. Rep.* 11: 24144



www.wikipedia.org

High diversity of modes of action

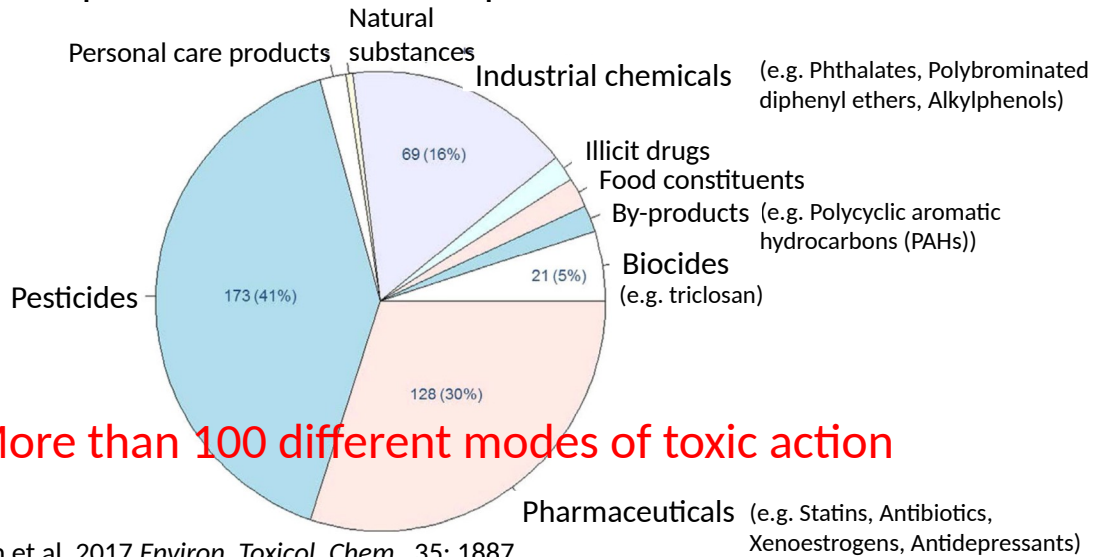
426 compounds found in 3 European river basins



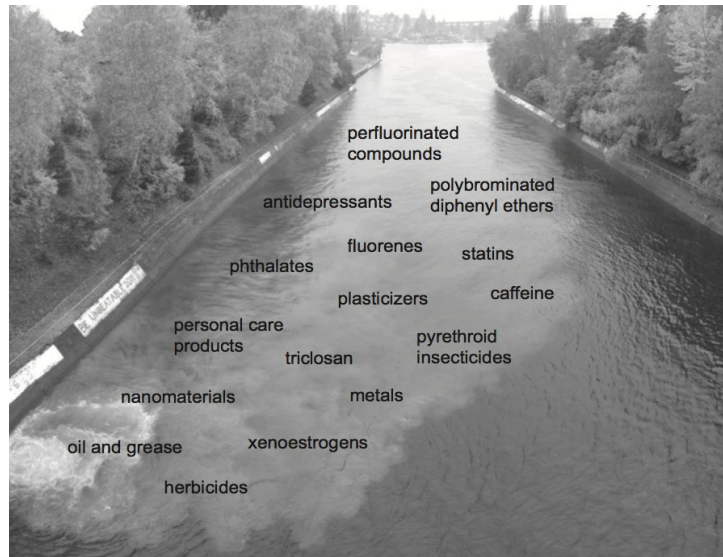
The study found 426 compounds, although the real number would be higher given that legacy chemicals such as PCBs or metals were not measured.

High diversity of modes of action

426 compounds found in 3 European river basins



How to assess the joint effect of chemicals?

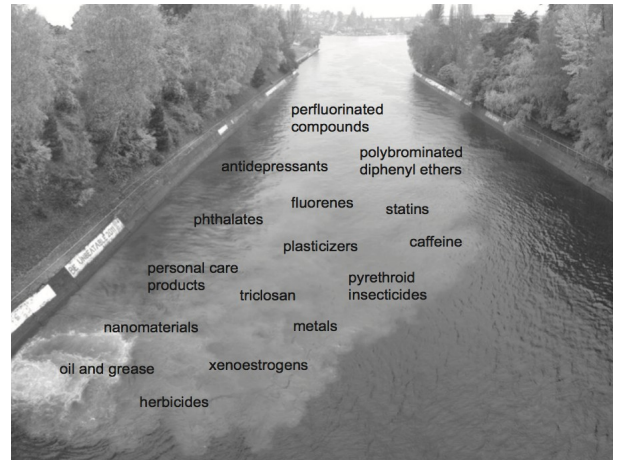


Amiard-Triquet, C., Amiard, J. C., & Mouneyrac, C. (Eds.). (2015). Aquatic ecotoxicology: Advancing tools for dealing with emerging risks. Elsevier/AP, Academic Press is an imprint of Elsevier.

How to assess the joint effect of chemicals?

Imagine that we detected 20 chemicals at concentrations $c_1, c_2 \dots c_{20}$. What data would you need as a prerequisite to estimate their joint effect and how could you calculate a joint effect?

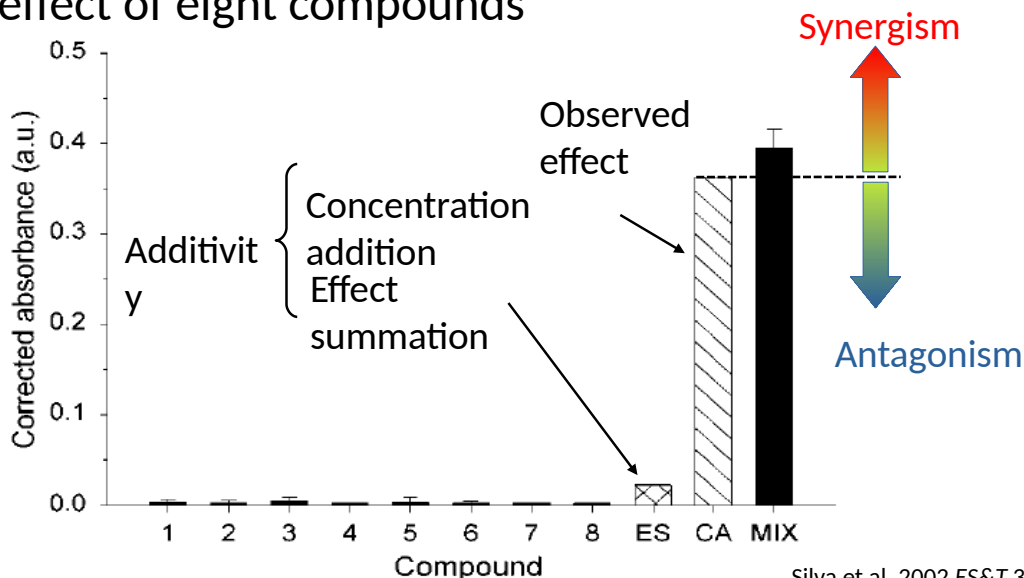
Discuss 5 minutes with your neighbour and then report to class.



Amiard-Triquet, C., Amiard, J. C., & Mouneyrac, C. (Eds.). (2015). Aquatic ecotoxicology: Advancing tools for dealing with emerging risks. Elsevier/AP, Academic Press is an imprint of Elsevier.

Mixture toxicity can matter!

Joint effect of eight compounds



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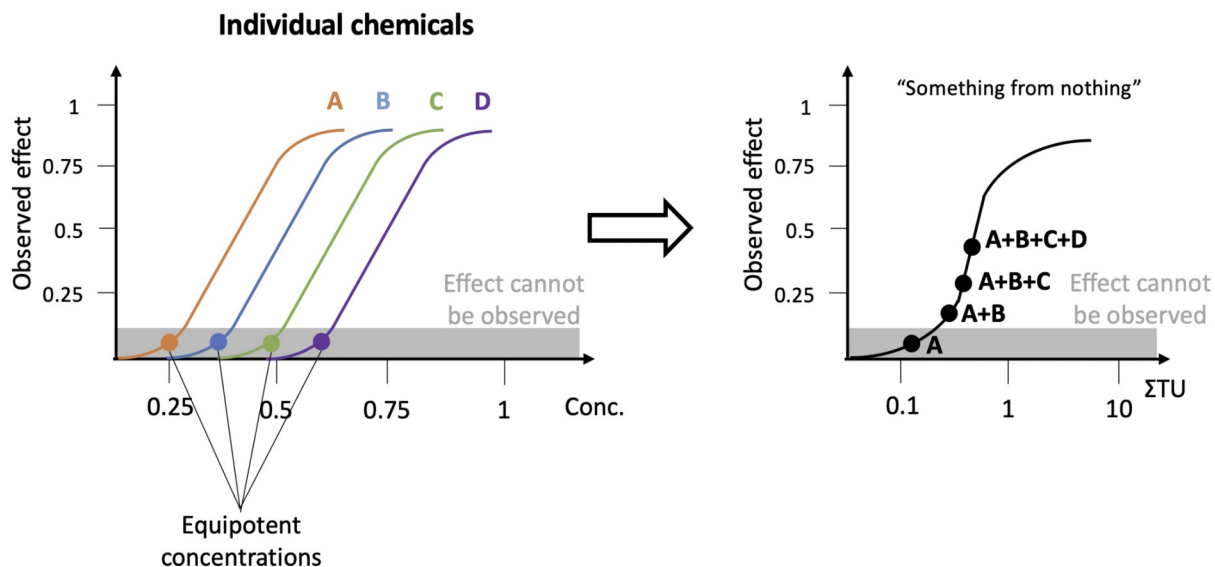
Effect summation is also termed Independent Action.

Antagonism: Mixture toxicity is smaller than expected (i.e. smaller than predicted by a null model, here the concentration addition model)

Synergism: Mixture toxicity is larger than expected (i.e. larger than predicted by a null model)

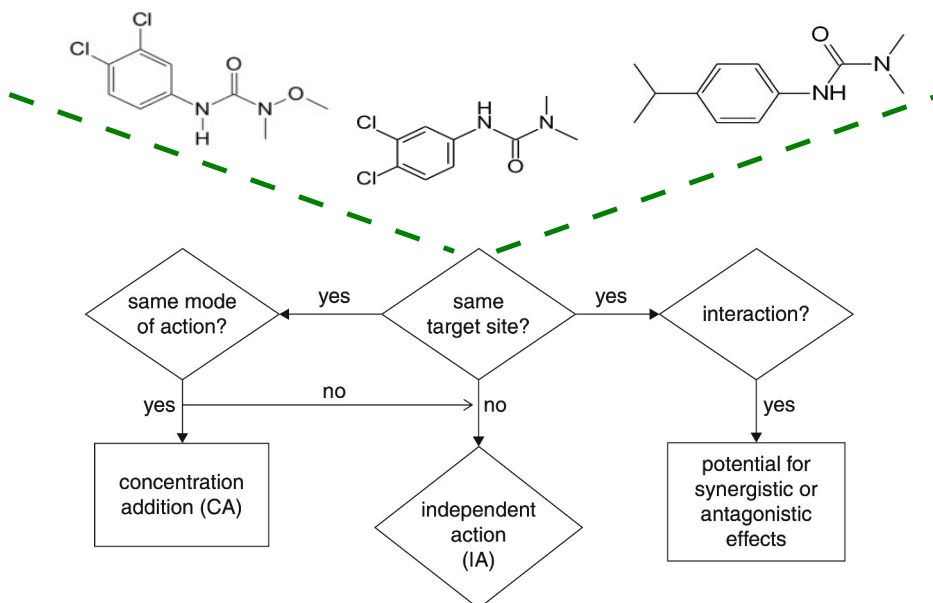
Additivity: Toxicity of mixture can be predicted by a null model, here CA or IA/ES.

Something from nothing effect



The *something from nothing* effect is only a consequence of adding up concentrations benchmarked by their toxicity, it does not imply mechanistic interactions.

Mixtures and mode of action



Escher B. 2013: Modes of Action of Chemical Pollutants. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg. DOI: 10.1007/SpringerReference_379983

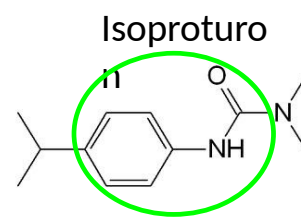
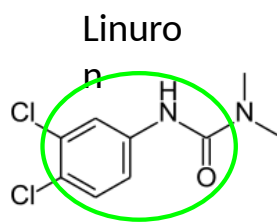
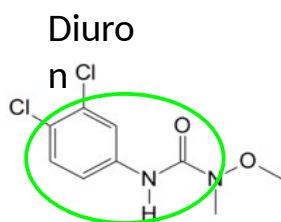
Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^n \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

c_i : concentration of compound i
 c_{mix} : concentration of the mixture
 ECx_i : effect concentration $x\%$ for i
 ECx_{mixCA} : mixture concentration having $x\%$ of effects following CA

Chevre & Gregoria
2013

Example: Phenylurea herbicides



Algae 72-h EC50,
growth ($\mu\text{g L}^{-1}$)

2.7

16

13

Chèvre N., Gregorio V. 2013: Mixture Effects in Ecotoxicology. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg. DOI: 10.1007/SpringerReference_379982

CA assumes that the compounds in the mixture have the same concentration-response relationship (same slope). A more flexible model represents Generalized CA that allows for different slopes, see:

Howard G.J. & Webster T.F. (2009) Generalized concentration addition: A method for examining mixtures containing partial agonists. Journal of Theoretical Biology 259, 469–477.

Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^n \frac{p_i}{EC_{xi}} \right)^{-1} \text{ with } p_i = \frac{c_i}{c_{mix}}$$

Observed conc. (µg L ⁻¹)	5.4	160	1.3
p_i	$\frac{5.4}{166.7} = 0.032$	$\frac{160}{166.7} = 0.96$	$\frac{1.3}{166.7} = 0.008$
Algae 72-h EC50, growth (µg L ⁻¹)	2.7	16	13
$\frac{p_i}{EC_{xi}}$	$\frac{0.032}{2.7} = 0.012$	$\frac{0.96}{16} = 0.06$	$\frac{0.008}{13} = 0.0006$

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$$(x)^{-1} = \frac{1}{x}$$

Concentration addition (CA)

$$ECx_{mixCA} = \left(\sum_{i=1}^n \frac{p_i}{EC_{xi}} \right)^{-1} \quad \text{with} \quad p_i = \frac{c_i}{c_{mix}}$$

$$\frac{p_i}{EC_{xi}} \quad \frac{0.032}{2.7} = 0.012 \quad \frac{0.96}{16} = 0.06 \quad \frac{0.008}{13} = 0.0006$$

$$\sum_{i=1}^n \frac{p_i}{EC_{xi}}$$

$$0.012 + 0.06 + 0.0006 = 0.0726$$

$$ECx_{mixCA} = \frac{1}{0.0726} = 13.77$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^n \frac{c_i}{EC_{xi}}$$

c_i : concentration of compound i
 EC_{xi} : effect concentration x% for compound i

Observed conc. ($\mu\text{g L}^{-1}$)	5.4	160	1.3
Algae 72-h EC50, growth ($\mu\text{g L}^{-1}$)	2.7	16	13
$\frac{c_i}{EC_{xi}}$	$\frac{5.4}{2.7} = 2$	$\frac{160}{16} = 10$	$\frac{1.3}{13} = 0.1$

$$sumTU = 2 + 10 + 0.1 = 12.1$$

CA and Toxic units (TU)

Sum of TU based on CA:

$$sumTU = \sum_{i=1}^n \frac{c_i}{EC_{xi}} \quad \begin{array}{l} c_i: \text{concentration of compound } i \\ EC_{xi}: \text{effect concentration } x\% \text{ for compound } i \end{array}$$

$$sumTU = 2 + 10 + 0.1 = 12.1$$

$$\text{Same as } \frac{\sum_{i=1}^n c_i}{ECx_{mixCA}} = \frac{166.7}{13.77} = 12.1$$

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)]$$

$E(c_{mixIA})$: predicted effect of the mixture following IA

$E_{(ci)}$: effect of compound i at concentration c

Example: Insecticide, herbicide and fungicide

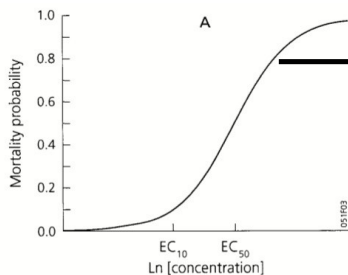
	Diuro	Thiacloprid	Tebuconazole
Observed concentrations ($\mu\text{g L}^{-1}$)	50	10	20

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)]$$

Example: Insecticide, herbicide and fungicide

	Diuro	Thiacloprid	Tebuconazole
Observed concentrations (µg L)	50	10	20



Derive $E(c_i)$

$E(c_i)$ 0.05 0.2 0.35

Independent action (IA)

$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)]$$

$$E(c_i) \quad 0.05 \quad 0.2 \quad 0.35$$

$$1 - E(c_i) \quad 0.95 \quad 0.8 \quad 0.65$$

$$\prod_{i=1}^n [1 - E(c_i)] \quad 0.95 * 0.8 * 0.65 = 0.49$$

$$E(c_{mixIA}) = 1 - 0.49 = 0.51$$

Computation of IA often hampered by data availability
(mostly only NOEC, LOEC, EC50 reported)

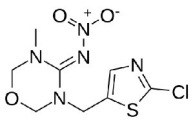
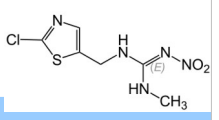
Chevre & Gregoria

Chèvre N., Gregorio V. 2013: Mixture Effects in Ecotoxicology. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg. DOI: 10.1007/SpringerReference_379982

Exercise

$$E(c_{mixIA}) = E(c_1 + c_2 + \dots + c_n) = 1 - \prod_{i=1}^n [1 - E(c_i)] \quad sumTU = \sum_{i=1}^n \frac{c_i}{EC_{xi}}$$

Please calculate the mixture toxicity using CA as sumTU and IA models for the following situation:

	Thiametoxam		Clothianidin
			
96-h LC50 Chironomus riparius (µg/L)	100		29
Observed concentrations (µg/L)	5		2.9
$E(c_i)$	0.02		0.2

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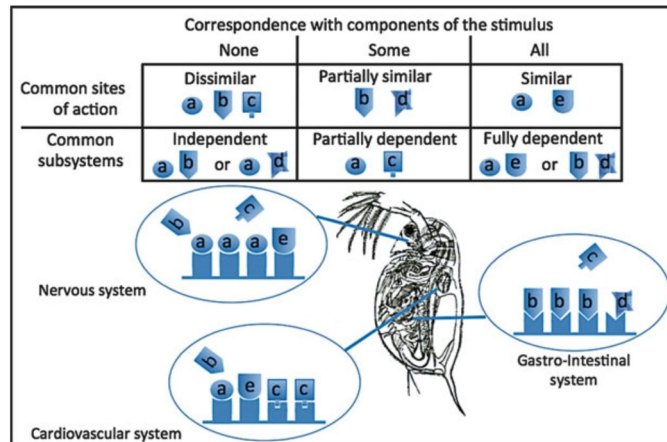
Result:

SumTU = 0.15

E(c mixIA) = 21.6%

CA or IA?

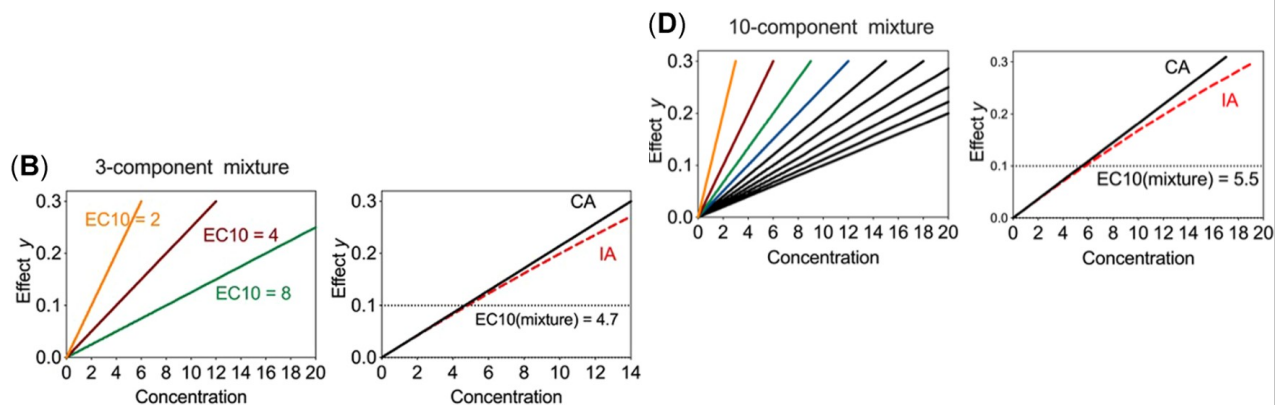
- Same target site and MoA → CA
- But MoA can vary between organs



Chèvre N., Gregorio V. 2013: Mixture Effects in Ecotoxicology. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg. DOI: 10.1007/SpringerReference_379982

CA or IA?

- For equipotent mixtures with low effect levels (i.e. $<EC_{10}$), often predictions converge
- CA standard because data often too limited for IA



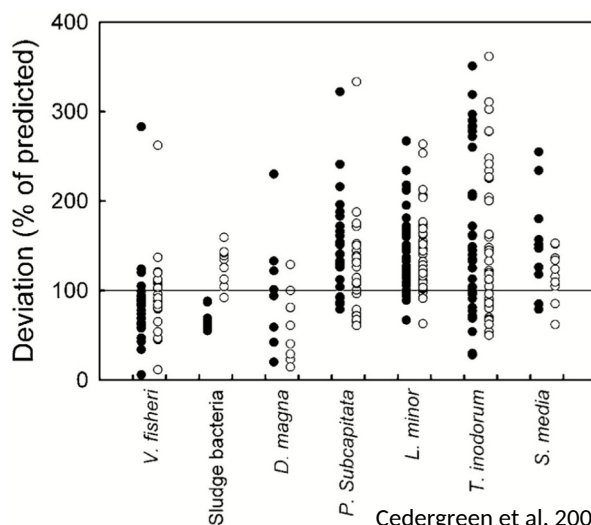
23

Escher et al. 2020 *Environ. Toxicol. Chem.*

Escher, B., Braun, G., & Zarfl, C. (2020). Exploring the Concepts of Concentration Addition and Independent Action Using a Linear Low Effect Mixture Model. *Environmental Toxicology and Chemistry*, 39(12), 2552–2559. <https://doi.org/10.1002/etc.4868>

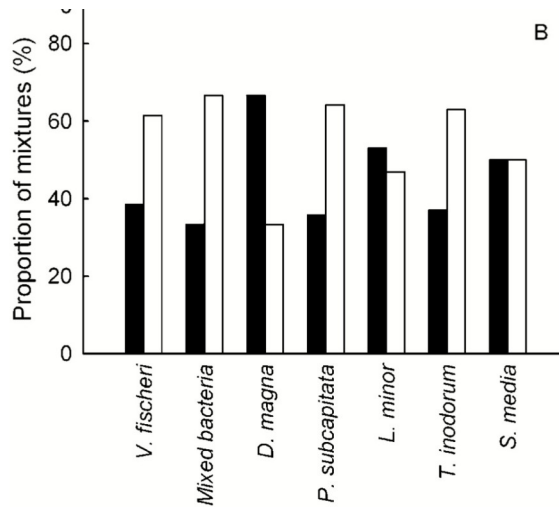
How good is the prediction?

Ratio of CA (black) and IA (white) prediction to observed EC50



How good is the prediction?

Proportion of mixtures better described with CA (black) and better described with IA (white)

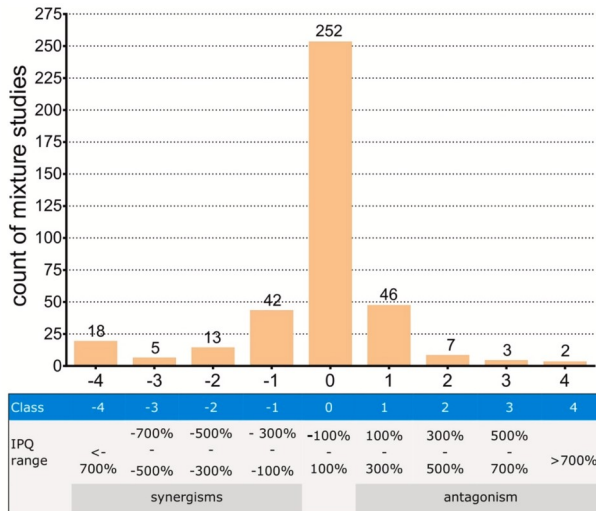


The results are based on 158 data sets.

There is slight indication that IA is better than CA (56 of 98 mixtures where better described by IA), but *Daphnia magna* shows the opposite trend.

How good is the prediction?

Observations largely within 1 order of magnitude of mixture prediction



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Martin et al. 2021 *Environ. Intern.* 146: 106206

IPQ = Index of Prediction Quality

The formula is given below, where the ratio is the observed to predicted effect. This means that if the observed effect concentration (equivalent to weaker effect) is twice the predicted effect concentration, the IPQ is (+)100%. Conversely, if the observed is half of the predicted effect concentration, the IPQ is -100%.

$$IPQ = \begin{cases} 100 * (Ratio - 1) \text{ if } Ratio \geq 1 \\ 100 * \left(1 - \frac{1}{Ratio}\right) \text{ if } Ratio < 1 \end{cases}$$

How to identify synergism and antagonism?

- Model Deviation Ratio (MDR) often used in Ecotoxicology:

$$\text{MDR} = \frac{\text{Expected/predicted EC with null model}}{\text{Observed EC}}$$

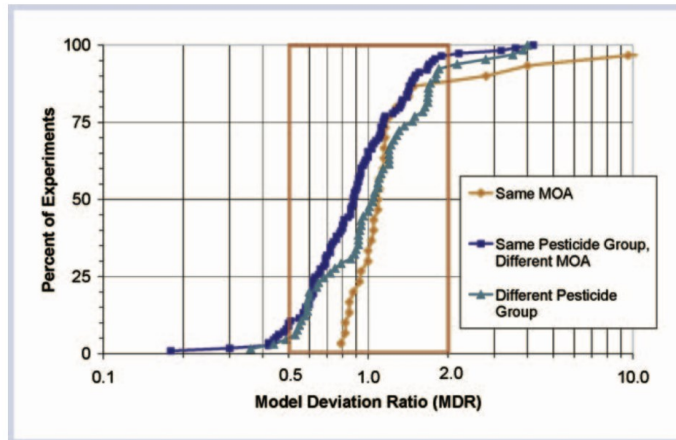
- MDR sets threshold based on effect size (e.g. 0.5 and 2)
- Why not statistical significance?
 - Sample-size dependent: e.g. low effect size (deviation from model) significant for large sample size, high effect size not significant for small sample size

Note that Belden et al. (2007) uses the reciprocal ratio, i.e. predicted to observed. Therefore, $\text{MDR} > 1$ means that the prediction underestimated the effect, i.e. greater toxicity. A value < 1 means that the prediction overestimated the effect.

Note also that if effects were compared directly, what is underestimation and overestimation may again swap.

How good is the prediction?

MDR for CA for pesticide mixtures → Reasonable match for majority of experiments, low fraction of synergism/antagonism



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Belden et al. 2007 *Integr. Environ. Assess. Manag.* 3 (3): 364

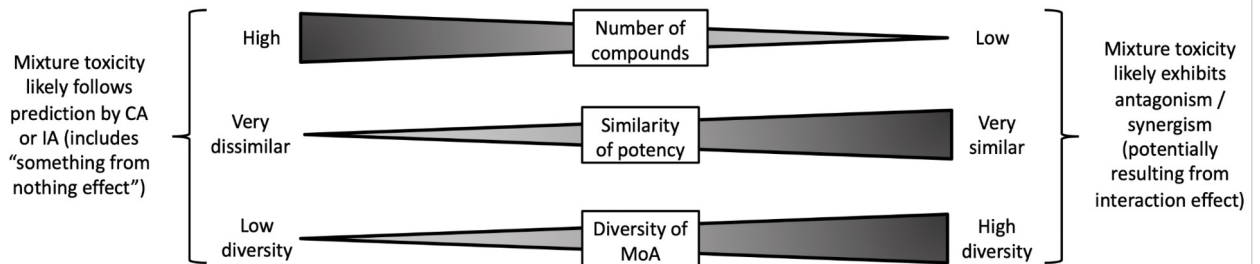
The figure shows the cumulative distribution of MDR for concentration addition experiments grouped by mixtures with the same mode of action ([MOA]; $n = 30$, indicated by diamond symbol), mixtures of the same pesticide use group and different MOA ($n = 113$, indicated by square symbol), and mixtures of different pesticide use groups and different MOA ($n = 64$, indicated by square symbol).

Species tested are shown in Table 1 of Belden et al. (2007)

Table 1. The number of experiments testing concentration addition (CA), independent action (IA), and simple interaction (SI) models for each taxonomic group. $n = 303$

Higher taxonomic group	Predictive model		
	CA	IA	SI
Amphibia	4	—	3
Bivalvia	2	—	—
Chlorophyta	87	10	—
Crustacea	20	4	14
Insecta	16	1	39
Liliopsida	38	19	—
Osteichthyes	38	1	3

How frequent are synergism and antagonism?



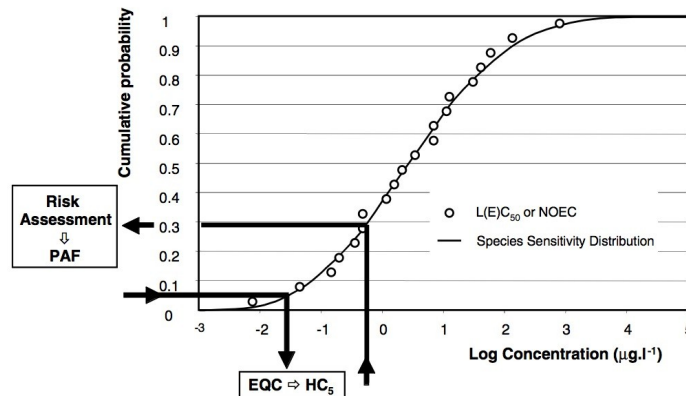
msPAF: Mixture prediction for SSDs

- Mixture prediction (CA and IA/RA) for community level effects
- Instead of ECx and $E(c)$ values, use Hcp and PAF values from single-substance SSDs

IA is termed response addition (RA) in this context

SSDs for risk assessment

- Hazardous concentration (HC_p): conc. at which $p\%$ affected
- Potentially affected fraction PAF: given by concentration



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Cutoff for community PNEC

Posthuma, Traas & Suter 2002

Posthuma, L., Traas, T. P., & Suter, G. W. (2002). General Introduction to species sensitivity distributions. In L. Posthuma, G. W. Suter, & T. P. Traas (Eds.), *Species sensitivity distributions in ecotoxicology* (pp. 3–10). Lewis.

msPAF: Mixture prediction for SSDs

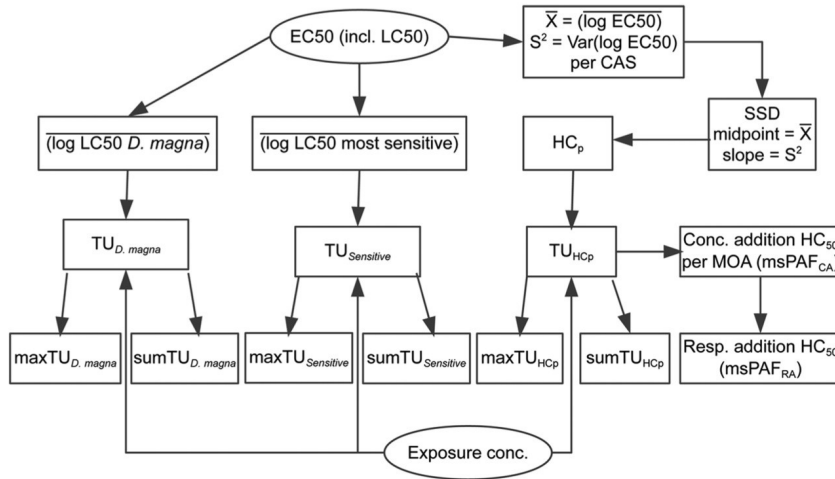
- Mixture prediction (CA and IA/RA) for community level effects
- Instead of ECx and $E(c)$ values, use Hcp and PAF values from single-substance SSDs
- $msPAF_{CA}$: Aggregate Hcp for all compounds with same MoA
- Alternative: $sumTU_{HCP}$

$$msPAF_{RA} = 1 - \prod_{i=1}^n (1 - msPAF_{CA,i})$$

IA is termed response addition (RA) in this context

Example for community-level mixture indices

IA/RA was only calculated in SSD context due to lack of data



Example is for freshwater invertebrate communities, in case of other organism groups, a different test organism than *Daphnia magna* would be relevant.

Example for community-level mixture indices

For 4 of 5 data sets $sumTU_{HCS}$ best metric, but maxTU and metrics based on *D. magna* or most sensitive test species not much worse

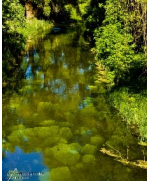
Exposure metric	Rank sum of metric across countries	Number of times metric was among models within a range of 2 to BIC of best-fit model
$sumTU_{HCS}$	14	4
$sumTU_{Sensitive}$	18	3
$maxTU_{HCS}$	21	3
$maxTU_{Sensitive}$	21	3
$maxTU_{D.magna}$	22	2
$sumTU_{D.magna}$	26	3
$msPAF_{RA}$	32	1
$maxTU_{HCS0}$	35	1
$sumTU_{HCS0}$	36	1

Example is for freshwater invertebrate communities, in case of other organism groups, a different test organism than *Daphnia magna* would be relevant.

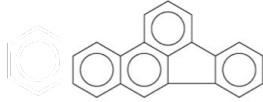


Chemicals don't occur in isolation

Analysis of german WFD monitoring data



Excessive nutrients



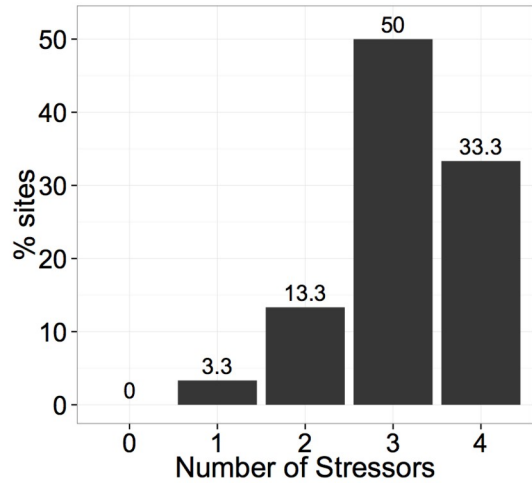
Toxicants



Invasive species



Habitat degradation

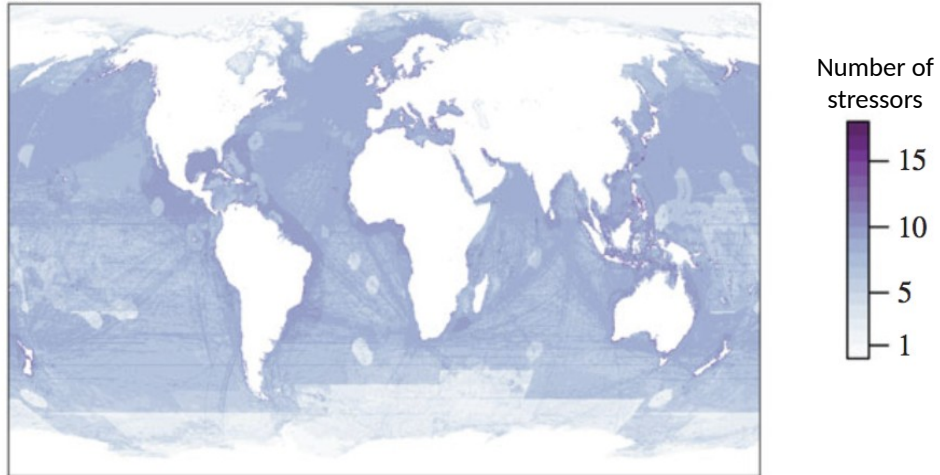


Schäfer et al. 2016 *Freshw. Biol.* 61: 2116

In more than 80% of sites at least 3 stressors occur above risk thresholds.

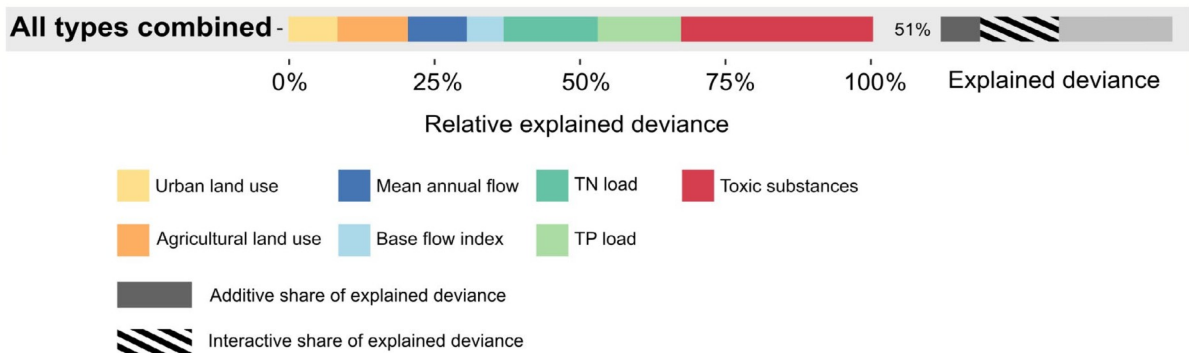
Chemicals don't occur in isolation

(a) Global distribution of marine stressors



Chemicals don't occur in isolation

- Analysis of European WFD monitoring data
- Several stressors/drivers of stressors explain variability in ecological status including mixtures
- Relevant fraction of interactions



A wide range of chemical mixtures
with different MOAs and multiple
non-chemical stressors co-occur
that can potentially affect
organisms and may interact

What are mechanisms underlying interactions?

Two stressors or chemicals can interact directly or indirectly regarding their (direct) effect

Direct interaction:

Stressor/Chemical A moderates intensity of Stressor/Chemical B

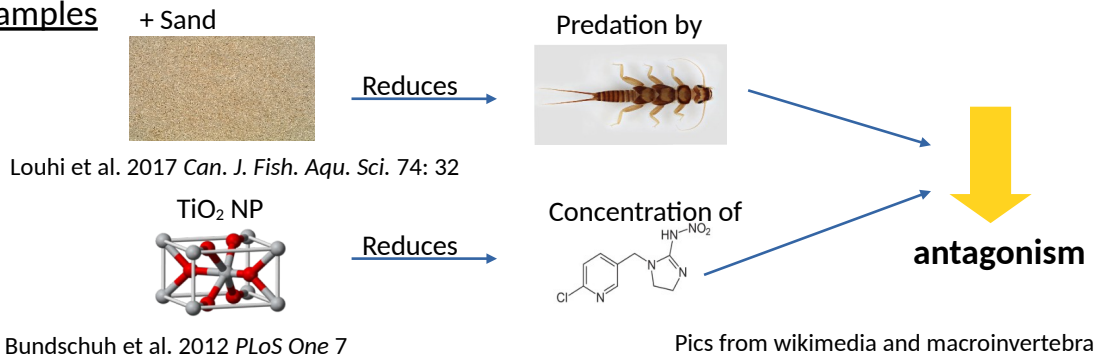
What are mechanisms underlying interactions?

Two stressors or chemicals can interact directly or indirectly regarding their (direct) effect

Direct interaction:

Stressor/Chemical A moderates intensity of Stressor/Chemical B

Examples



What are mechanisms underlying interactions?

Two stressors or chemicals can interact directly or indirectly regarding their (direct) effect

Indirect interaction:

Stressor/Chemical A moderates sensitivity to Stressor/Chemical B

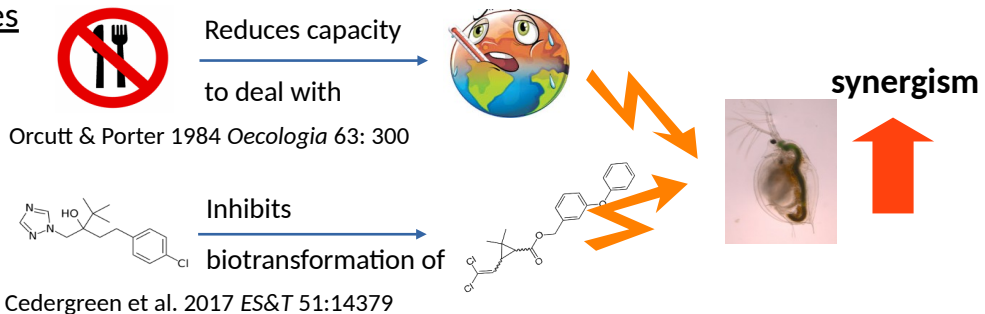
What are mechanisms underlying interactions?

Two stressors or chemicals can interact directly or indirectly regarding their (direct) effect

Indirect interaction:

Stressor/Chemical A moderates sensitivity to Stressor/Chemical B

Examples



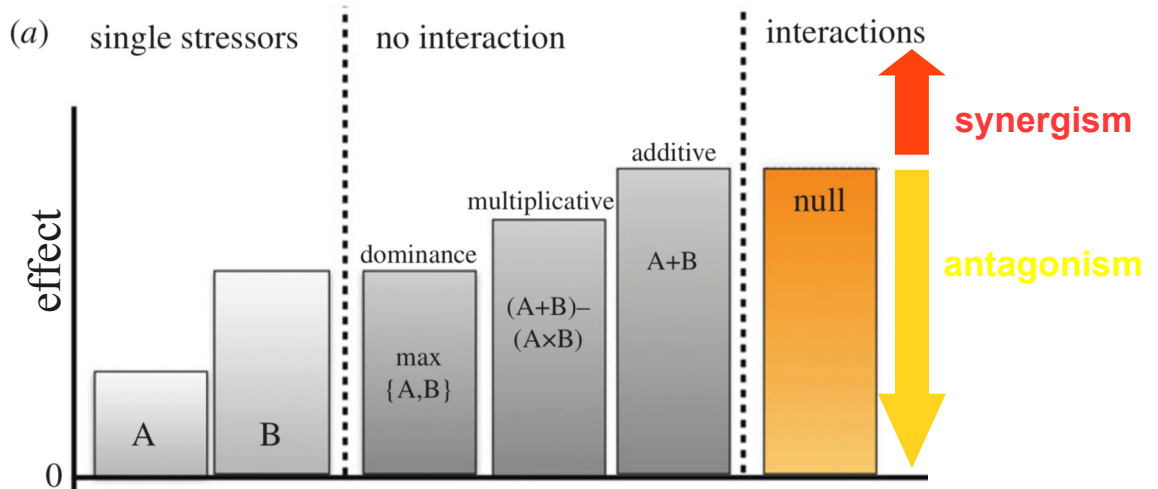
How frequent are synergism and antagonism?

Meta-analysis of 15 meta-analyses on multiple stressors:

- between 4% and 68% synergistic responses
- few consistent responses (no antagonism on physiological level, no synergism on community- or ecosystem level)

Synergism and antagonism in multiple stressor research

Interaction = (statistically significant) deviation from null model

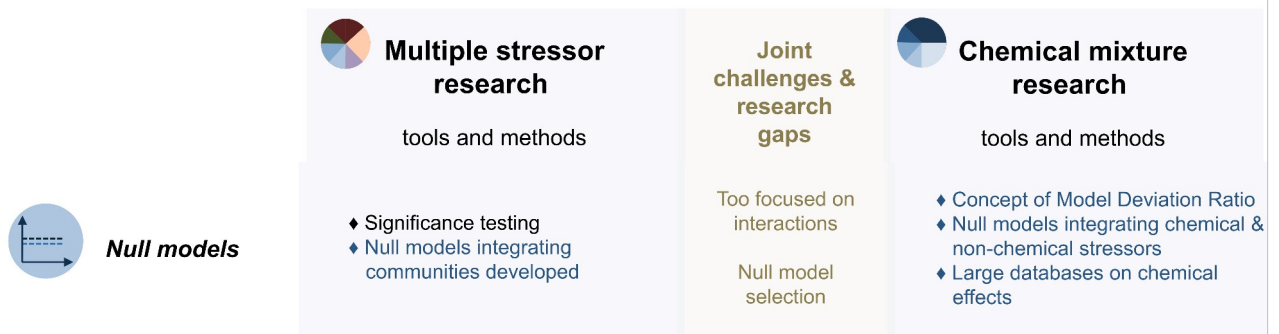


Joint challenge, different approaches

- Multiple stressor/ ecological and ecotoxicological null models have different null model standards
- Novel null model developed to integrate chemical and non-chemical stressors

Ecological model	(Eco)toxicological model	Null model equation for multiple stressor research	Required input data
Simple addition	–	$f_{AB}(Sl_A, Sl_B) = f_A(Sl_A) + f_B(Sl_B)$	Effects for A and B
Multiplicative	Effect addition	$f_{AB}(Sl_A, Sl_B) = f_A(Sl_A) + f_B(Sl_B) - f_A(Sl_A) f_B(Sl_B)$	Effects for A and B as proportions (%) ^b
Dominance	–	$f_{AB}(Sl_A, Sl_B) = \max(f_A(Sl_A), f_B(Sl_B))$	Effects for A and B
–	Concentration addition ^a (CA)	$f_{AB}(Sl_A, Sl_B) = f_A(Sl_A + \gamma Sl_B)$ with $\gamma = Sl_{X_A}/Sl_{X_B}$	Stressor–effect relationship for A, stressor intensity for A and B and stressor intensities Sl_{X_A} and Sl_{X_B} that result in the same effect size x ($x = f_A(Sl_{X_A}) = f_B(Sl_{X_B})$), where x is typically selected as half the effect limit
–	Stressor addition (SAM)	$f_{AB}(Sl_A, Sl_B) = F_{Strcap}^{-1}(F_{Strcap}^{-1}(f_A(Sl_A)) + F_{Strcap}^{-1}(f_B(Sl_B)))$	Generalized density function for the distribution of the stress capacity f_{Strcap} ^c and effects for A and B ($f_A(Sl_A)$ and $f_B(Sl_B)$)

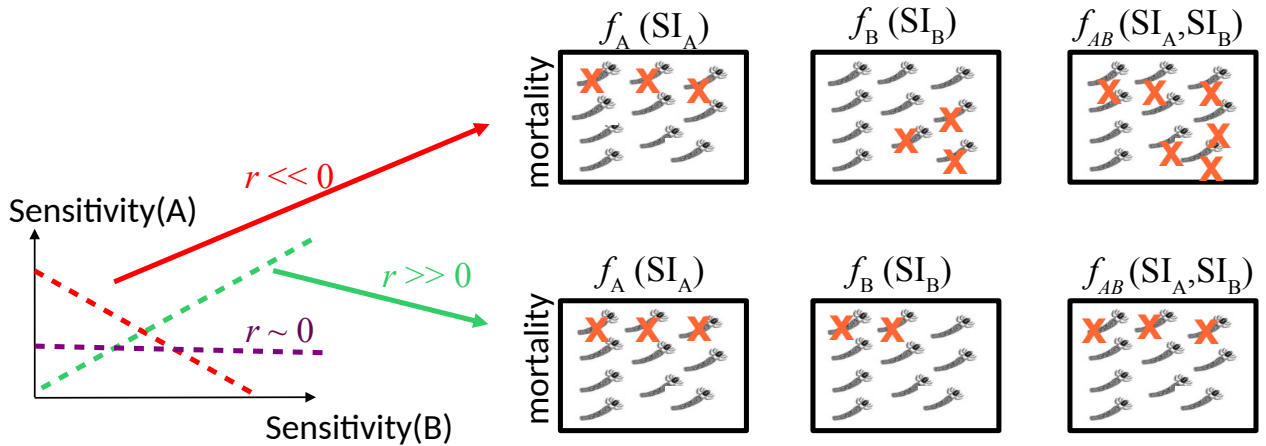
Joint challenge, different approaches



In case you are interested in the differences and similarities between multiple stressor research and chemical mixture research as well as to integrate both, have a look into the paper.

How to choose a null model?

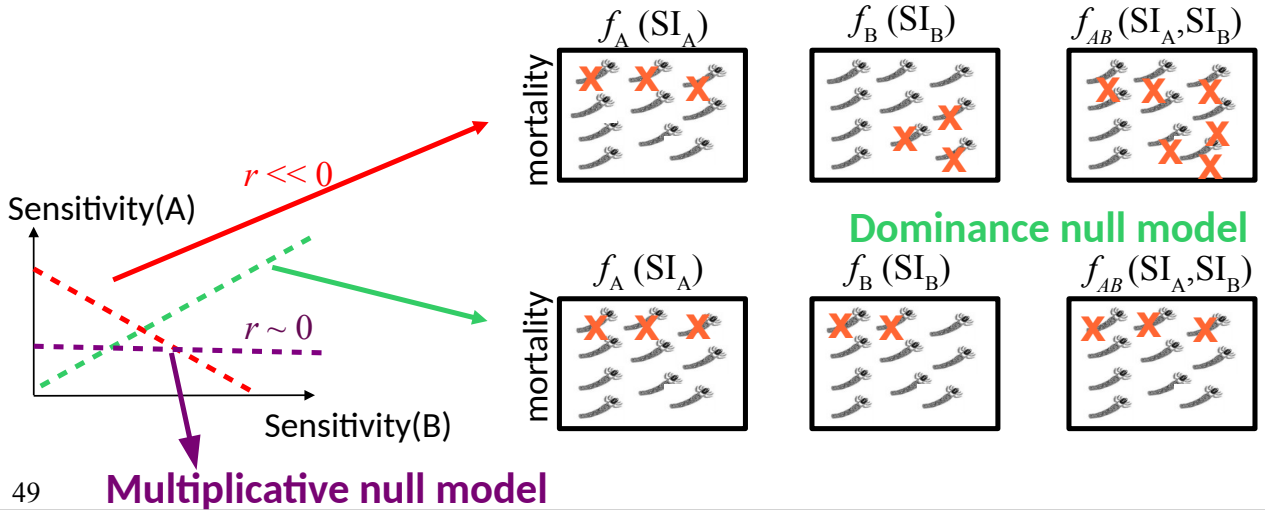
Correlations among stressor sensitivities may help



How to choose a null model?

Correlations among stressor sensitivities may help

Additive null model

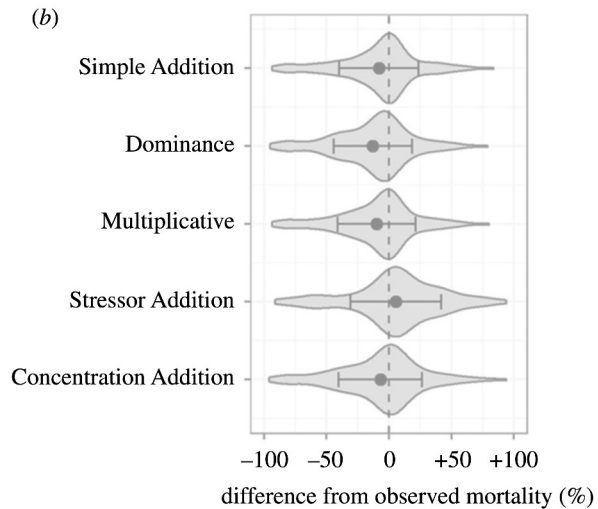


How to choose a null model?

- Stressor mode of action: How do the stressors affect the organisms? (e.g. physically, specific or non-specific physiologically) → Similar or different mode of action? Stressor interactions known?
- Sensitivities: Relationship between sensitivities to stressors?
- Effect type: What is affected? (e.g. mortality rather multiplicative, growth rather additive)
- Effect size: What are the individual effect sizes? (e.g. close to limit or differs by orders of magnitude → dominance)
- Knowledge of stressor-effect relationship: Broader range of models available

Null model comparison

SAM (and Concentration Addition model) least biased in meta-analysis

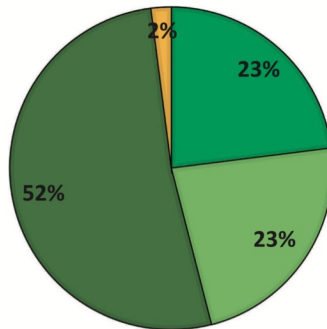


Null model comparison

Dominance best for warming and 2nd stressor (incl. contamination)

(d) Empirical results

■ Additive ■ Multiplicative
■ Dominance ■ Undefined



Larger than
predicted
by all null
models

35
%

Range
of null
model
predictions

23
%

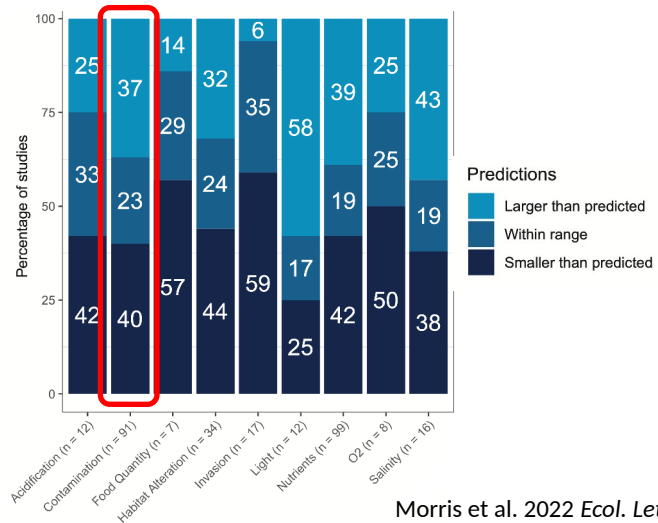
Smaller
than
predicted
by all null
models

42
%

How frequent are synergism and antagonism?

Quite frequent even if considering range of null models

Meta-analysis of studies on warming and 2nd stressor



Morris et al. 2022 *Ecol. Lett.* 25: 2540



Illustration of null models

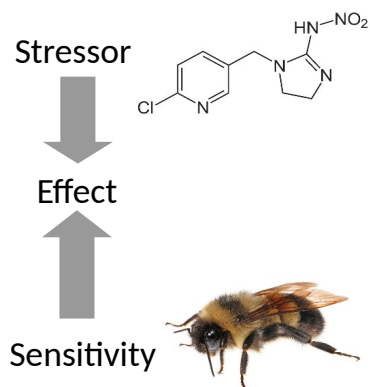


Illustration of null models: Standard models

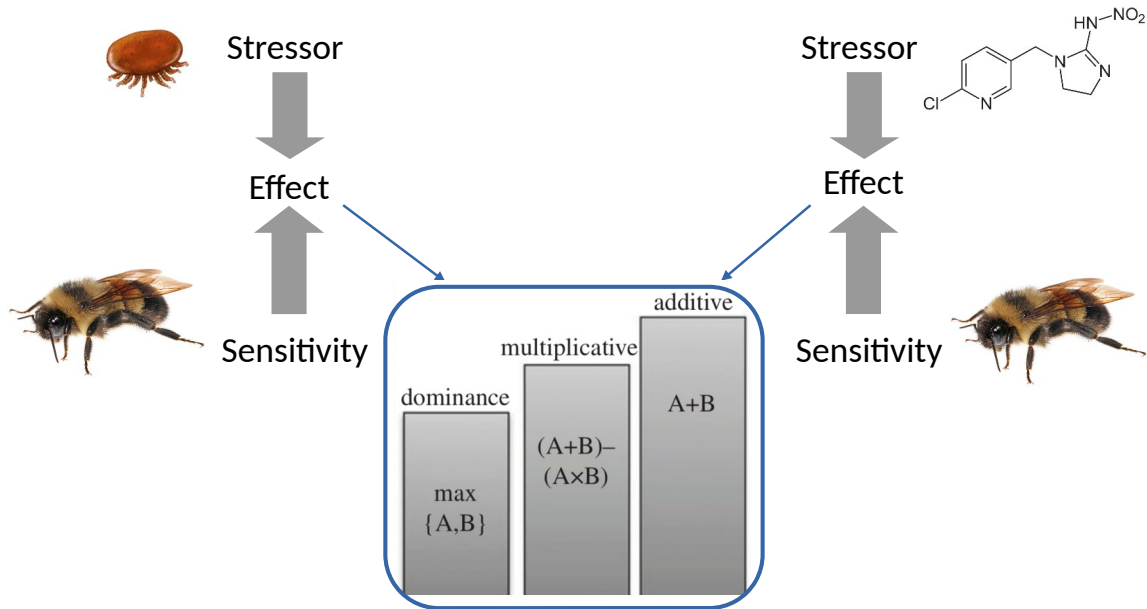
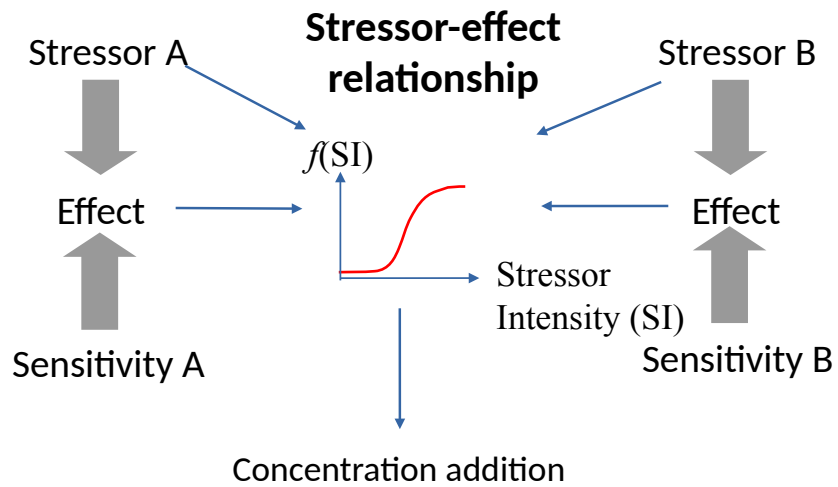
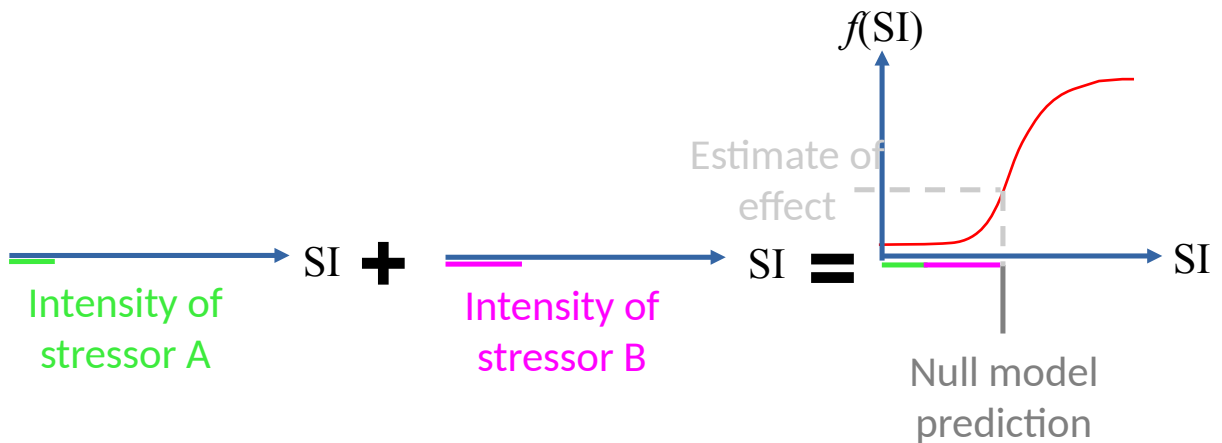


Illustration of null models: Concentration addition



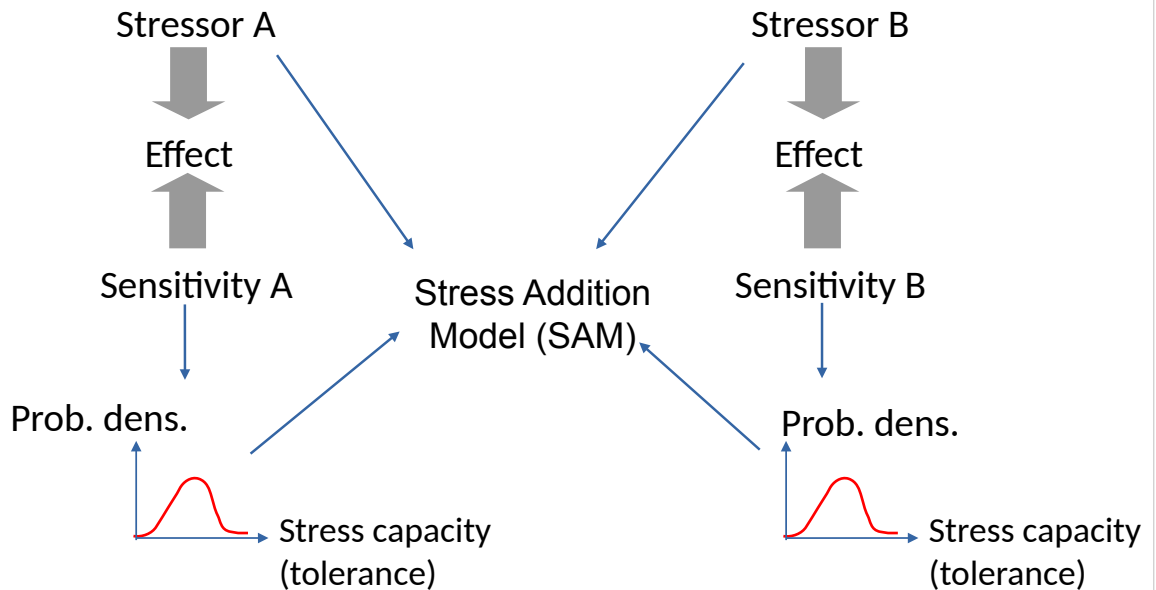
Concentration addition: Adding up stressor intensities



Note that not every stressor intensity is expressed as a concentration, but the term 'concentration addition' model is still used to avoid confusion with the stressor addition model described next.

CA does not directly predict an effect, it predicts a stressor intensity, which is related to an effect (e.g. EC_x for a mixture). If the TU approach is used, the sum of TUs is predicted, from which an effect could be estimated, if the relationship between TU and effect is known.

Illustration of null models: Stressor addition



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R code to run SAM is available associated with the publication of Dey & Koops (2021): https://github.com/cjdey/null_models_for_multiple_stressors

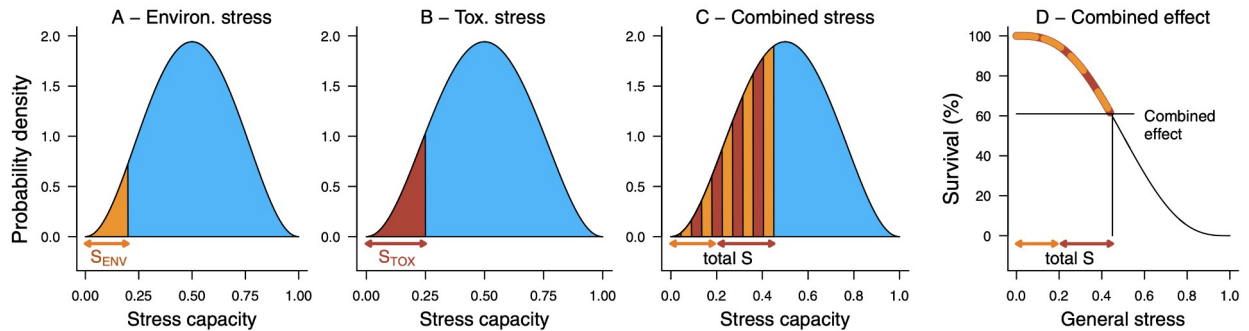
A related R package focussing on tri-phasic relationships is also available:

<https://git.ufz.de/oekotox/stressaddition>

SAM can also be applied through this webtool:

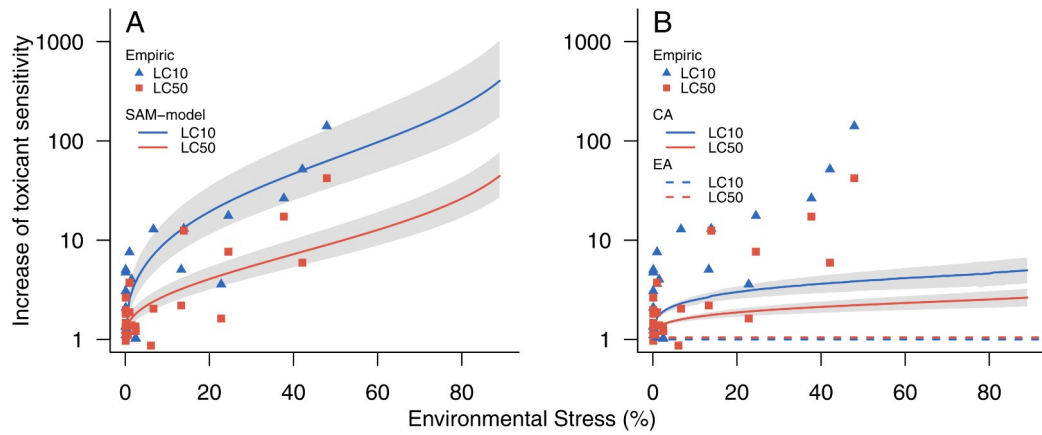
<https://www.systemecology.de/indicate/>

Stressor addition: Adding up stressor capacities

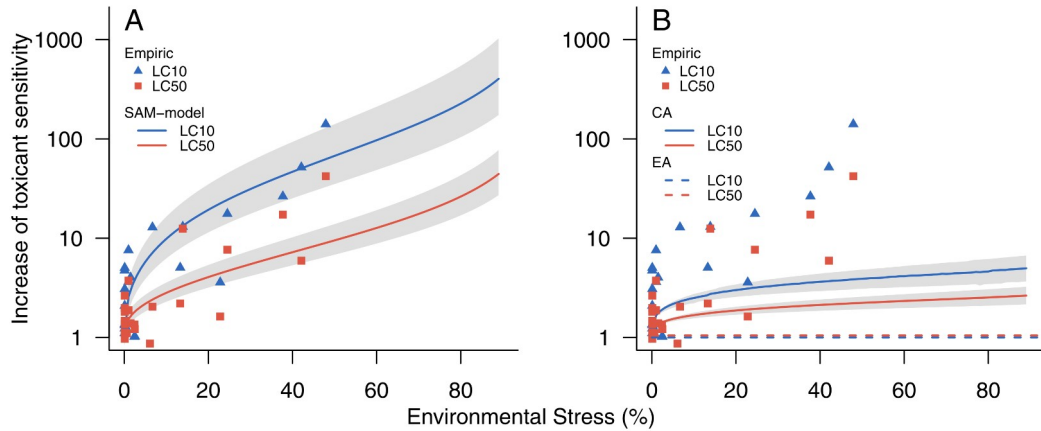


Liess M., Foit K., Knillmann S., Schäfer R.B. & Liess H.-D. (2016) Predicting the synergy of multiple stress effects. *Scientific Reports* 6, 32965.

Stressor addition: Adding up stressor capacities



Stressor addition: Adding up stressor capacities



SAM and CA rarely considered in multiple stressor/ ecological studies, because large majority of studies too few treatment levels

→ Use regression design

Griffin et al. 2016 *Mar. Ecol. Progr. Ser.* 543: 273,

Orr et al. 2024 *Ecol. Lett.* 27: e14463

Liess et al. 2016 *Sci. Rep.* 6, 32965